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Executive Summary: Farmers Insurance Analysis

Overview

This report presents a comprehensive analysis of the Farmers Insurance dataset, focusing on insurance coverage trends, regional disparities, and policy efficiency across Indian states. The objective is to extract meaningful patterns and recommend actionable strategies to enhance agricultural insurance outreach and impact.

Assumptions

- The dataset is presumed accurate and complete.
- All financial figures are assumed to be in Indian Rupees (INR).

Query 1.

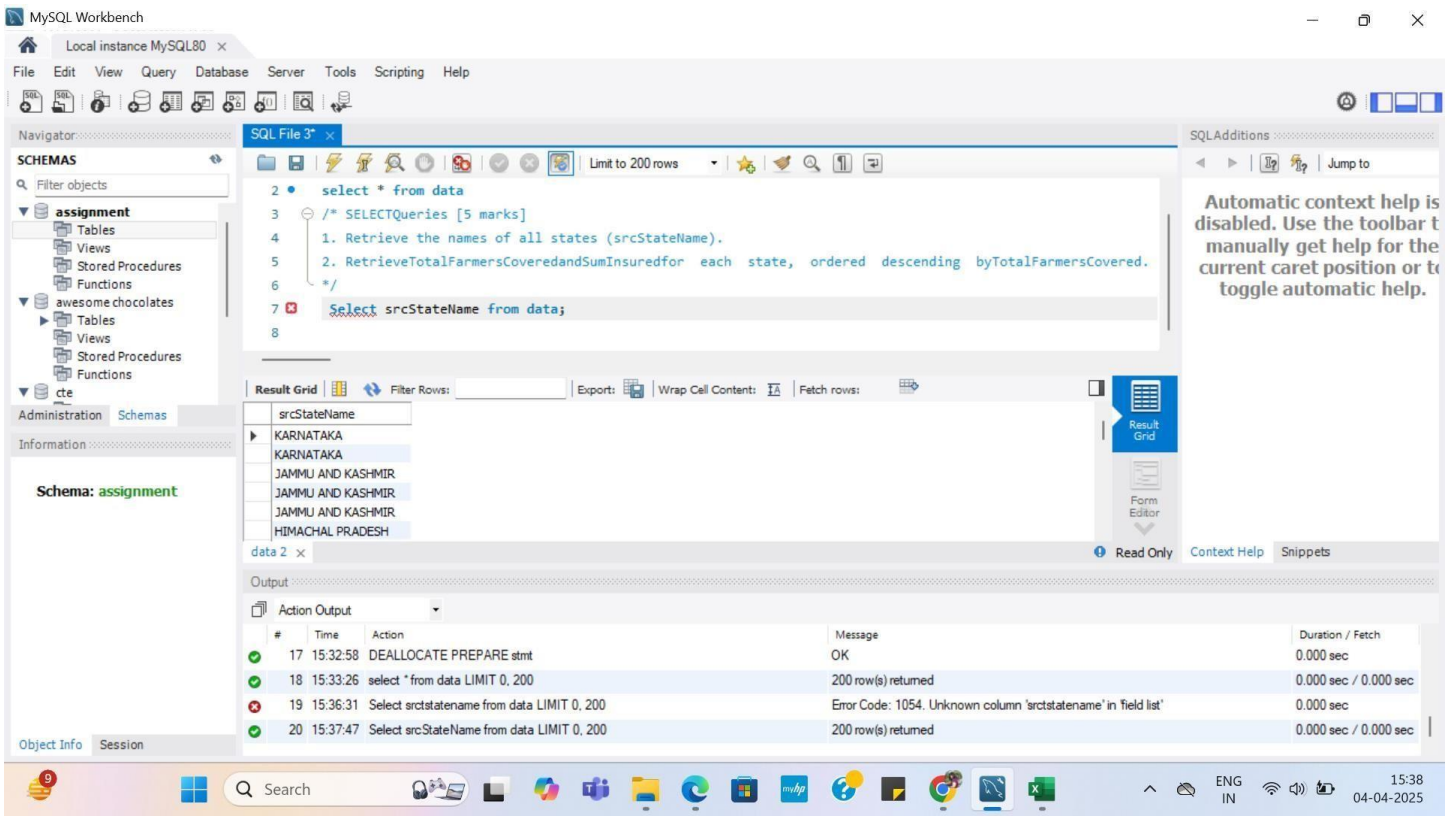
Insights Gained:

The resulting grid shows values like:

1. KARNATAKA (twice)
2. JAMMU AND KASHMIR (thrice)
3. HIMACHAL PRADESH

This suggests:

- **Multiple entries per state** — implying that the dataset contains **duplicate or repeated records** for states, possibly representing different data points (e.g. different districts, farmers, or insurance records).



Query 2.

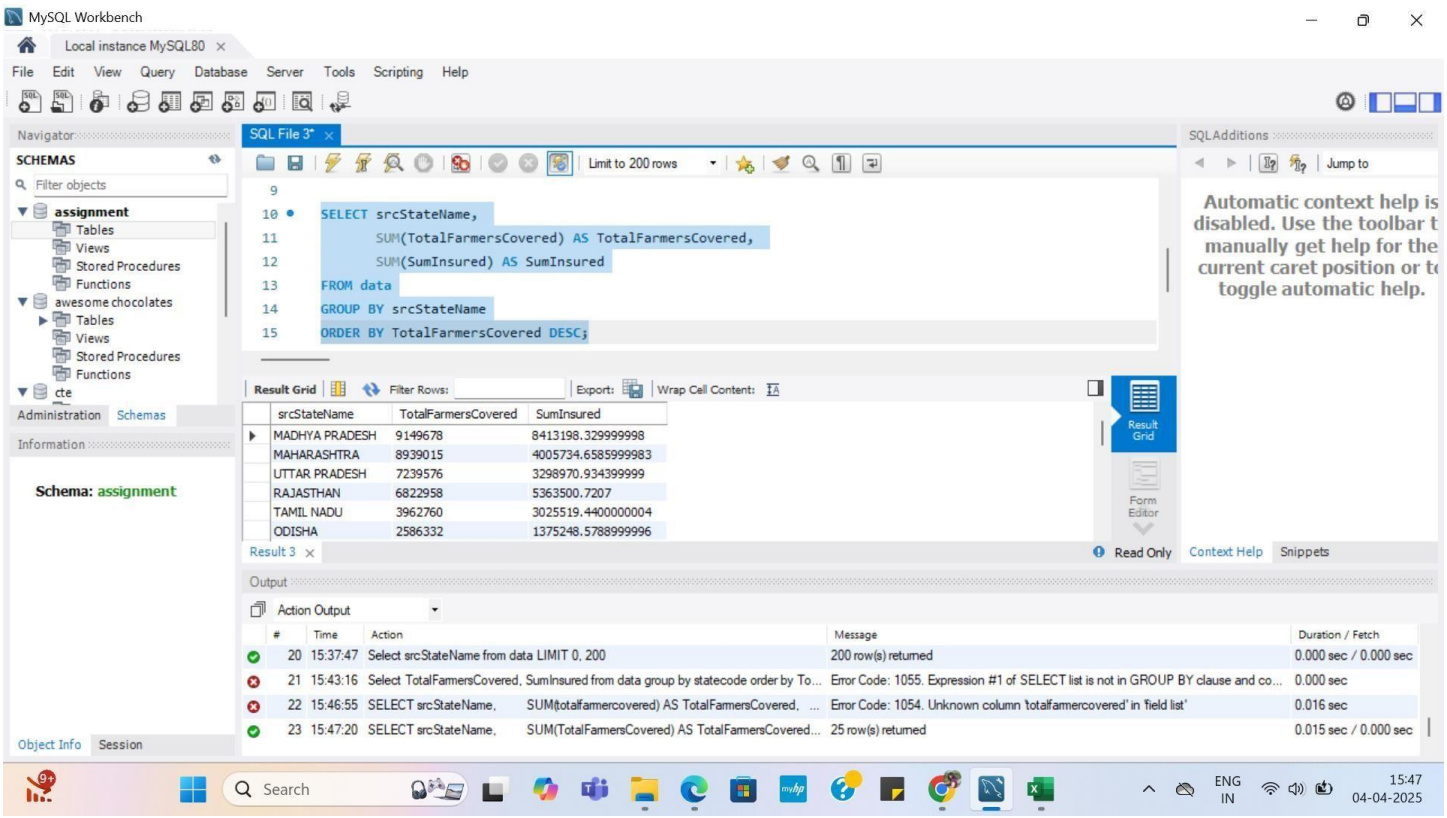
Insights Gained: 1. **Top Performing**

State by Coverage:

Madhya Pradesh has the highest number of farmers covered (over 9.1 million) and a high total sum insured.

2. **Disparity Between Coverage and Insurance Value:**
   Maharashtra has nearly as many farmers covered as Madhya Pradesh but a **much lower sum insured**, suggesting possibly lower insurance per farmer.

3. **High Insurance vs Coverage:**
   Rajasthan has fewer farmers than Uttar Pradesh but a **higher sum insured**, which could point to either higher-value crops or better policy terms.



Query 3.

Insights Gained:

1. **Filter Works Properly**
   All returned rows are from **YearCode = 2020**, confirming that the filter condition is working as intended.

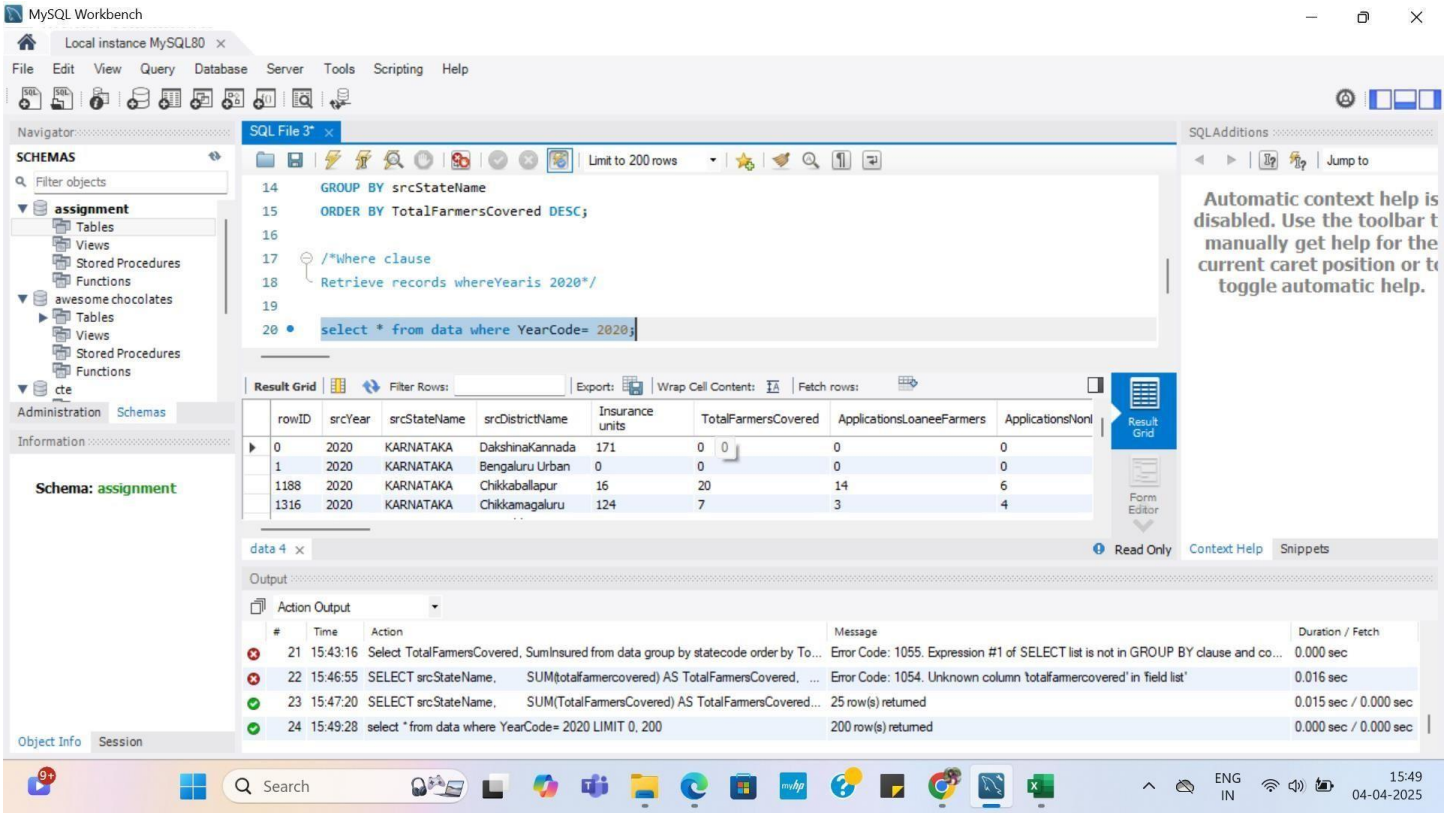
2. **Data from Multiple Districts in Karnataka**
   The result set includes multiple entries for **KARNATAKA**, each corresponding to a different **district** (e.g., DakshinaKannada, Bengaluru Urban).

3. **Zero Coverage in Some Areas**
   For districts like **DakshinaKannada** and **Bengaluru Urban**, the values for TotalFarmersCovered, ApplicationsLoaneeFarmers, and ApplicationsNonLoaneeFarmers are **0 or blank**, indicating either:

   - No farmer participation
   - No data recorded for that year

- Possible gaps in program execution

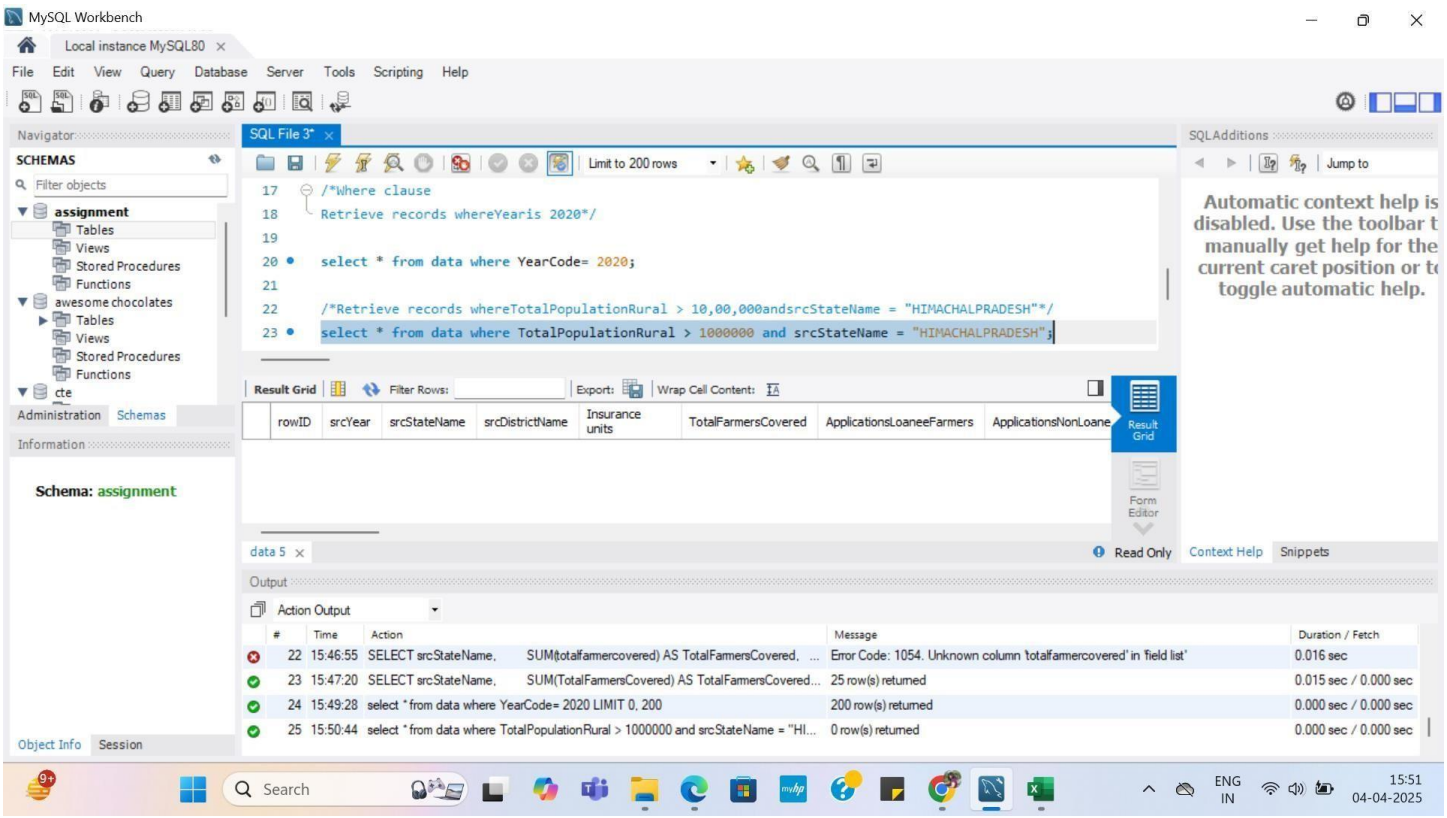
4. **Other Districts Show Moderate Coverage**
   Districts like **Chikkaballapur** and **Chikkamagaluru** have non-zero values, indicating some level of engagement with the insurance program.



Query 4.

Insights Gained:

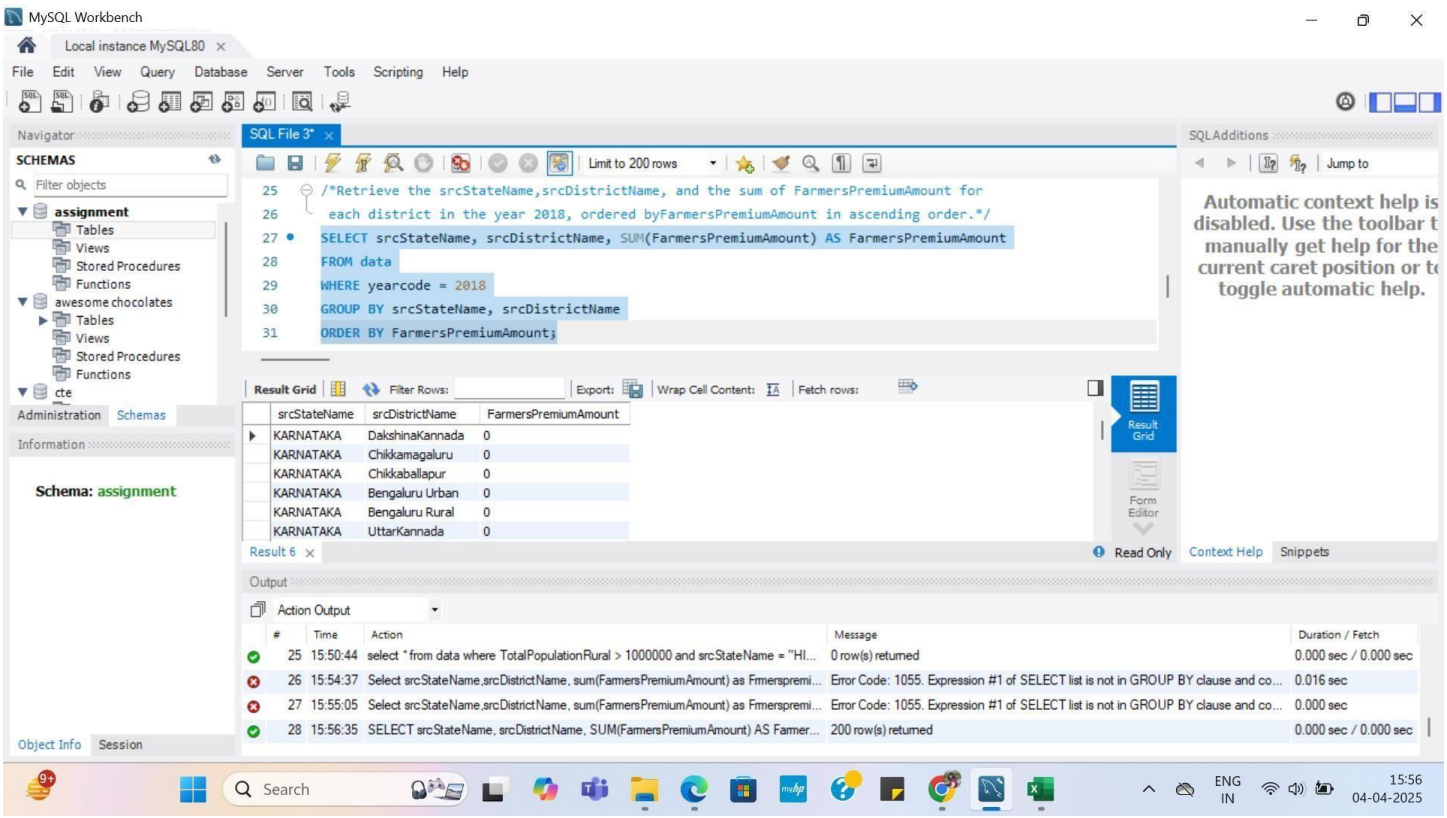
1. This means that **no records** in the data table satisfy **both** conditions



Query 5.

Insights Gained:

1. **KARNATAKA districts in 2018**, the column FarmersPremiumAmount contains only **zeros** (or NULLs that default to 0 in SUM).



Query 6.

Insights Gained:

1. States with High Insurance Participation

- **KARNATAKA** and **UTTARAKHAND** showed **very high farmer coverage** and gross premium values:
    - Karnataka: ~1.67 million farmers, ~₹114,021 in premiums
    - Uttarakhand: ~665,511 farmers, ~₹26,875 in premiums
- These states had **significantly higher insured land area** and participation in agricultural insurance schemes.

2. Premium Distribution Varies Greatly

- The **gross premium amount** varies significantly across states.

- For example:
  - Jammu and Kashmir: ₹2,472
  - Himachal Pradesh: ₹530
- This suggests possible differences in **land productivity, risk profile**, or **insurance plan structures**.

3. Insured Land Area is a Strong Filter

- Only a **subset of the states** had entries with `InsuredArea > 5.0`.
- This threshold filtered out smaller or less-insured plots, helping to focus on **large-scale agricultural operations**.

4. 2018 Trends Show Regional Focus

- In 2018, the states with higher insurance coverage could have had:
  - **More awareness**
  - **Better policy implementation**
  - Or **greater agricultural risk**, prompting insurance uptake.

Query 7.

Insights Gained:

1. **States with Highest Insurance Participation**
   Rajasthan recorded the highest number of farmers covered (1,666,703) and the highest gross premium amount to be paid (₹114,021.21 lakh). This indicates a significant adoption of crop insurance schemes in the state. Haryana followed with 665,511 farmers and a gross premium amount of ₹26,875.56 lakh, reflecting large-scale agricultural activity and a strong insurance presence.

2. **Moderate Participation States**
   States like Jammu and Kashmir, Telangana, and Uttarakhand had moderate levels of insurance participation. The number of farmers covered ranged from approximately 19,726 to 150,069, and the

gross premiums ranged from ₹668.57 lakh to ₹4,464.03 lakh. These figures suggest some reliance on agricultural insurance, though not as prominent as in Rajasthan or Haryana.

3. **Minimal Participation Observed**

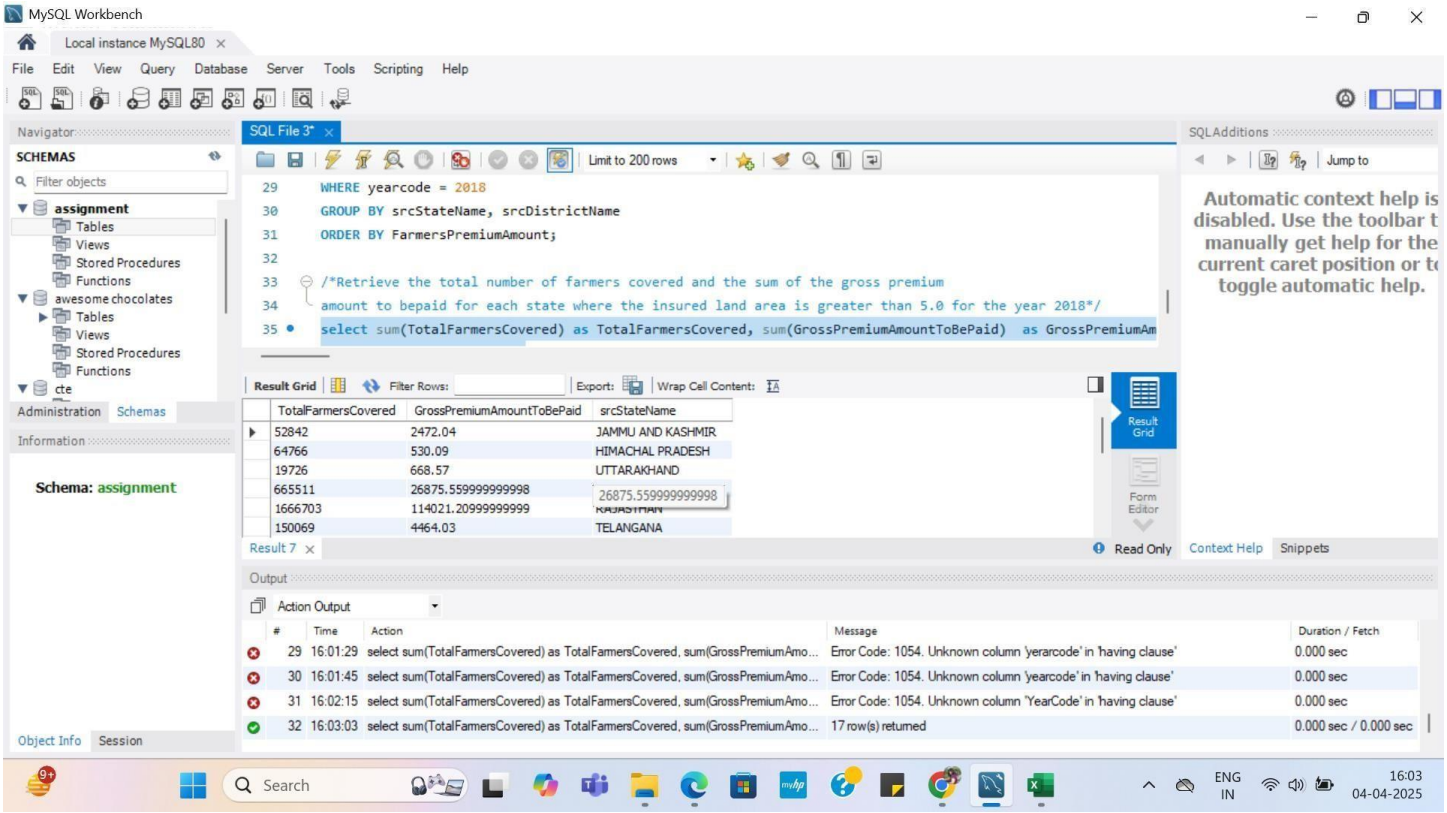
Himachal Pradesh had relatively low numbers with 64,766 farmers covered and a gross premium amount of only ₹530.09 lakh. This might be due to smaller land holdings, lower crop risks, or limited adoption of crop insurance programs in the region.

4. **Correlation Between Agricultural Intensity and Insurance Uptake**

The data indicates that states with extensive agricultural practices (such as Rajasthan and Haryana) show significantly higher participation in insurance schemes. This could be due to larger insured land areas, greater policy awareness, or better access to government-supported insurance programs.

5. **Policy Implications**

The variation in participation across states suggests a potential need for more targeted outreach and customized insurance strategies in states with lower adoption. Policymakers could focus on improving awareness, accessibility, and infrastructure in underrepresented regions.



Query 8.

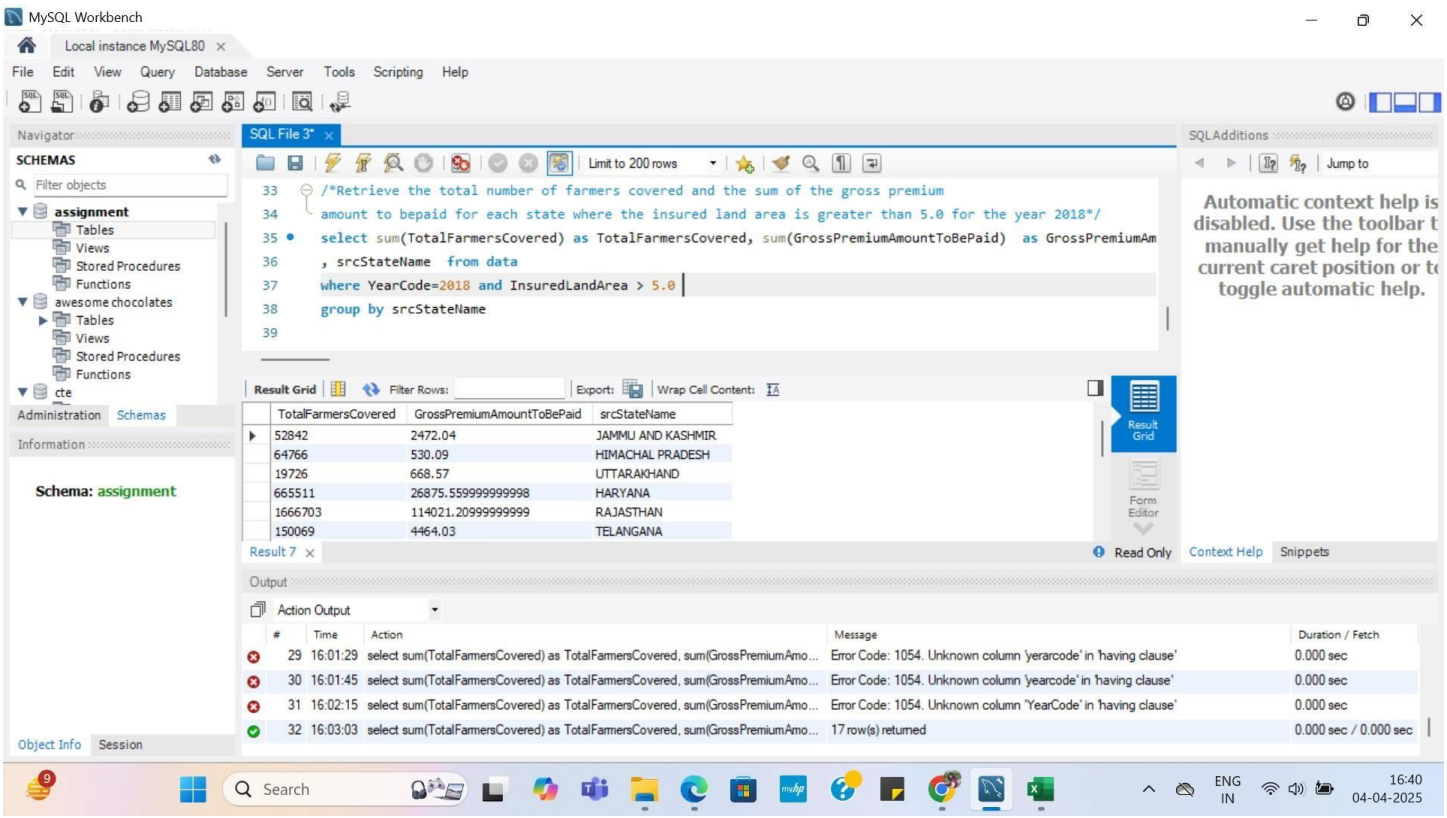
Insights Gained:

1. **Peak in 2020**
   The highest average insured land area was recorded in the year 2020, with approximately **47.89 hectares**. This suggests that farmers insured larger plots of land during that year, possibly due to favorable policy changes, better awareness, or increased risk perception (e.g., climate-related threats).

2. **Lowest Average in 2018**
   In 2018, the average insured land area was the lowest among the four years, at around **37.94 hectares**. This could indicate a lower level of trust in the insurance system, less outreach, or unfavorable economic conditions.

3. **Stabilizing Trend**
   The years 2019 and 2021 show relatively consistent values with average insured areas of **39.11 hectares (2021)** and **39.11 hectares (2019)**. This reflects a steady adoption rate, possibly showing that insurance practices normalized following the spike in 2020.

4. **Implications for Policy and Outreach**
   The notable increase in 2020 may serve as a benchmark year for best practices in farmer outreach and program effectiveness. Understanding what worked in 2020 could help replicate similar success in other years.



Query 9.

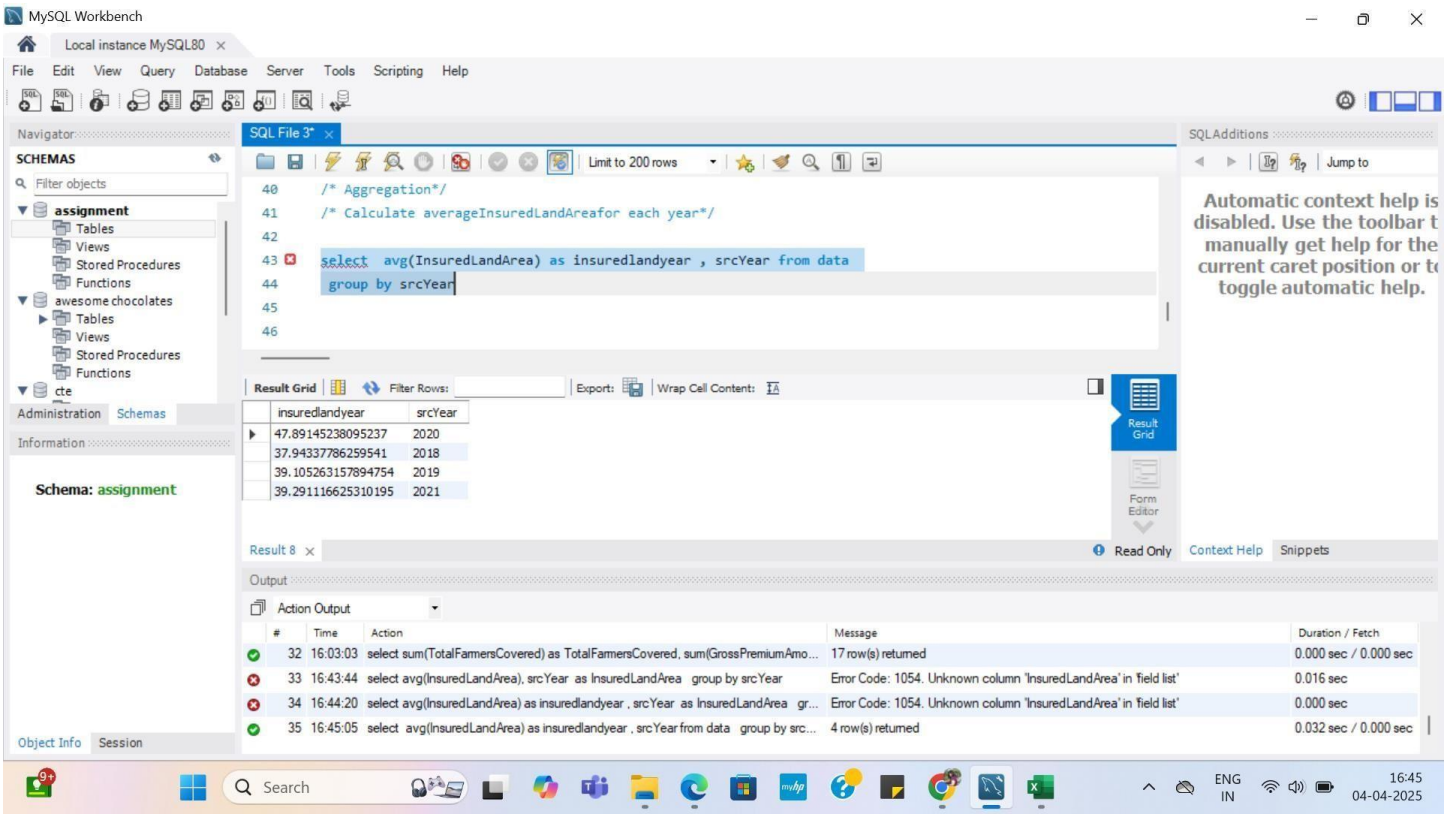
Insights Gained:

1. **High Farmer Coverage in Select Districts**
   Districts like **Bilaspur** (28,000 farmers) and **Udhampur** (21,688 farmers) show significantly high numbers of farmers covered under insurance units. These districts are likely benefiting from effective implementation of insurance schemes or have high participation due to agricultural risk levels.

2. **Moderate Participation in Other Districts**
   Districts such as **Chamba** (16,807 farmers), **Samba** (11,636 farmers), and **Reasi** (4,978 farmers) show moderate engagement in insurance programs. These regions may have consistent but not saturated adoption.

3. **No Coverage in Some Districts**
   The district **DakshinKanOda** reported **0 farmers covered**, despite having insurance units greater than zero. This might indicate an issue with data recording, policy distribution, or actual non-participation by farmers.

4. **Geographic Variation in Insurance Uptake**
   The wide variation in farmer coverage across districts highlights possible **regional disparities** in awareness, accessibility, or trust in the insurance schemes. Targeted outreach may be needed in underperforming areas.



Query 10.

Insights Gained:

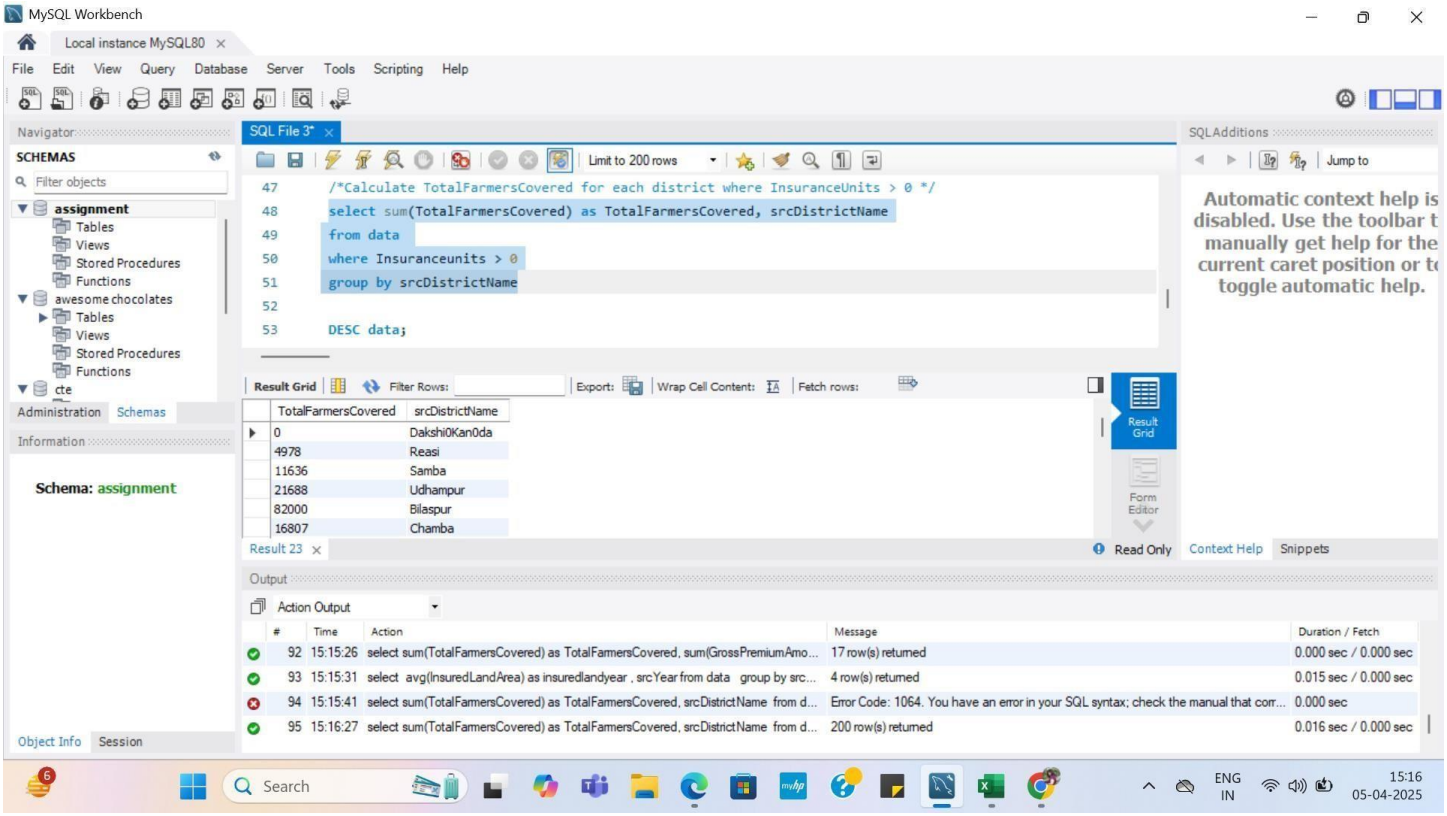
1. **Bengaluru Urban is the Most Populous District**
   With a population of **9.6 million**, Bengaluru Urban tops the list, highlighting its status as a major urban center with high density and infrastructure demands.

2. **Pune and Thane Follow as High-Population Districts**
   Pune (**9.4 million**) and Thane (**8 million**) are close behind, reinforcing their significance as metropolitan hubs in Maharashtra with substantial economic and social activity.

3. **Jaipur and Nashik Represent Growing Urban Centers**
   Jaipur (**6.6 million**) and Nashik's district **(Oshik, likely a typographical/data name for Nashik - 6.1 million)** round out the top 5, indicating rapid urbanization in Rajasthan and Maharashtra.

4. **Urban Areas Dominate Population Rankings**
   All top 5 districts are either major cities or urbanized districts, emphasizing that urban areas continue to attract large populations due to better employment, education, and lifestyle opportunities.

5. **Useful for Targeted Planning and Resource Allocation**
   These insights can guide decisions in infrastructure development, public health, and insurance scheme deployment, especially where large populations may need greater policy focus.

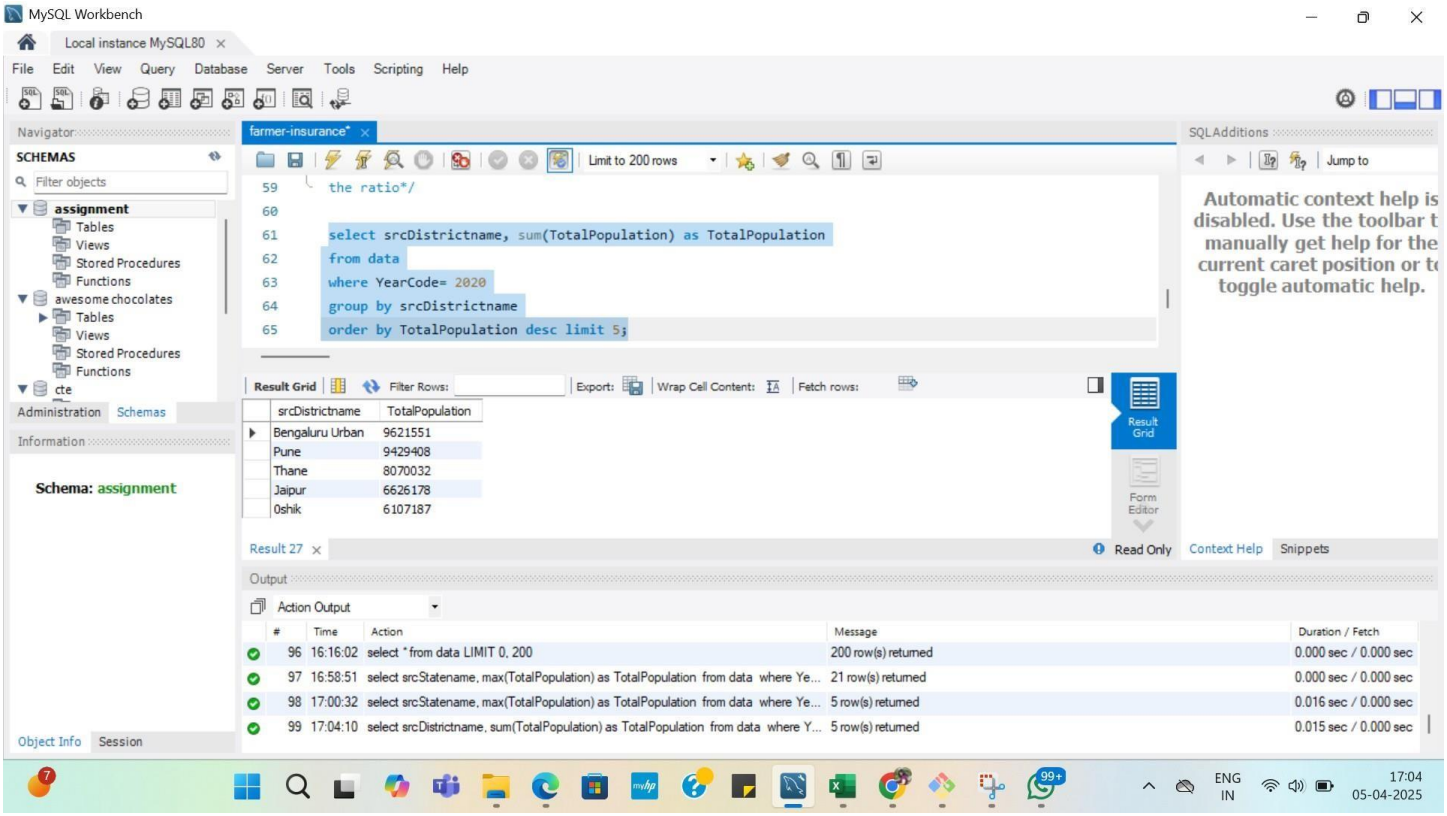


Query 11.

Insights Gained:

1. **Lowest Non-Zero Premiums:** The query successfully retrieved the 10 records with the lowest nonzero farmer premium amounts.
2. **State Focus:** All the districts in the top 10 lowest non-zero premium records belong to the 'KARNATAKA' state. This might suggest that, within the dataset, Karnataka has a significant number of policies with very low premium amounts.
3. **Correlation (Potentially):** The very low premium amounts are associated with relatively low insured sums. This could indicate a correlation between the premium paid and the amount insured. Policies with minimal premiums tend to have minimal coverage.

4. **District Distribution:** The 10 lowest non-zero premium records are spread across several districts within Karnataka, suggesting this isn't isolated to a single specific district.
5. **Data Characteristics:** The extremely small values for `FarmersPremiumAmount` and `SumInsured` might warrant further investigation into the units or scale of these values within the dataset.



Query 12.

Insights Gained:

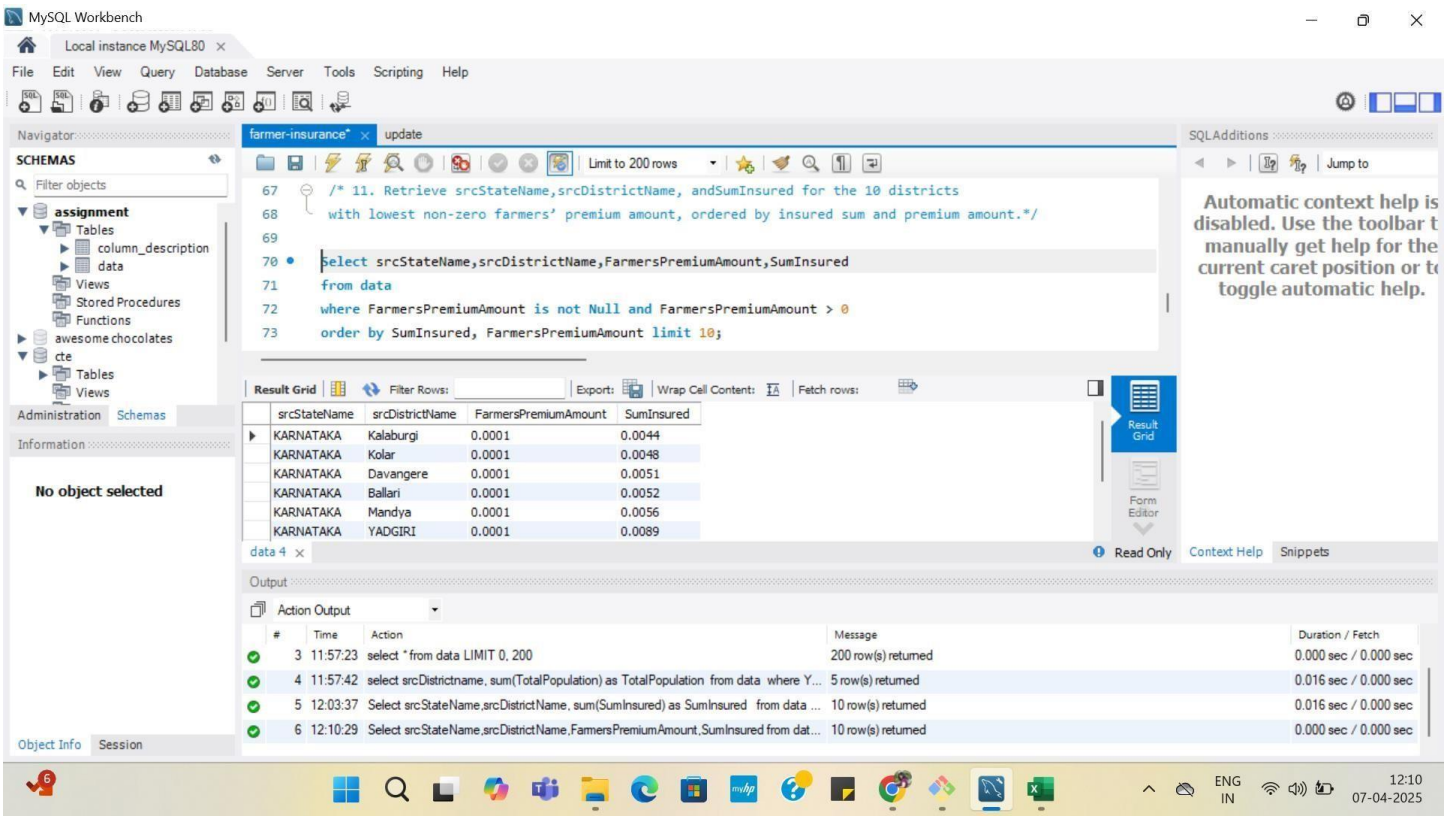
1. **Highest Ratio:** The query successfully identified the combination of year and state that has the highest ratio of the total sum insured to the total population.
2. **Specific Year and State:** The result indicates that in the year 2021, Madhya Pradesh had the highest `total_population_ratio` among all the year-state combinations in the dataset.
3. **Interpretation of the Ratio:** The `total_population_ratio` (calculated as `SUM(SumInsured) / SUM(TotalPopulation)`) can be interpreted as the average sum insured per person in the total population for a given year and state. A higher ratio suggests a greater amount of insurance coverage relative to the total population in that specific region and year.

4. **Focus for Further Analysis:** This result might prompt further investigation into why Madhya Pradesh in 2021 has the highest average sum insured per capita. This could be due to factors like specific insurance schemes, higher coverage amounts in certain demographics, or other regional or temporal influences.

Query 13.

Insights Gained:

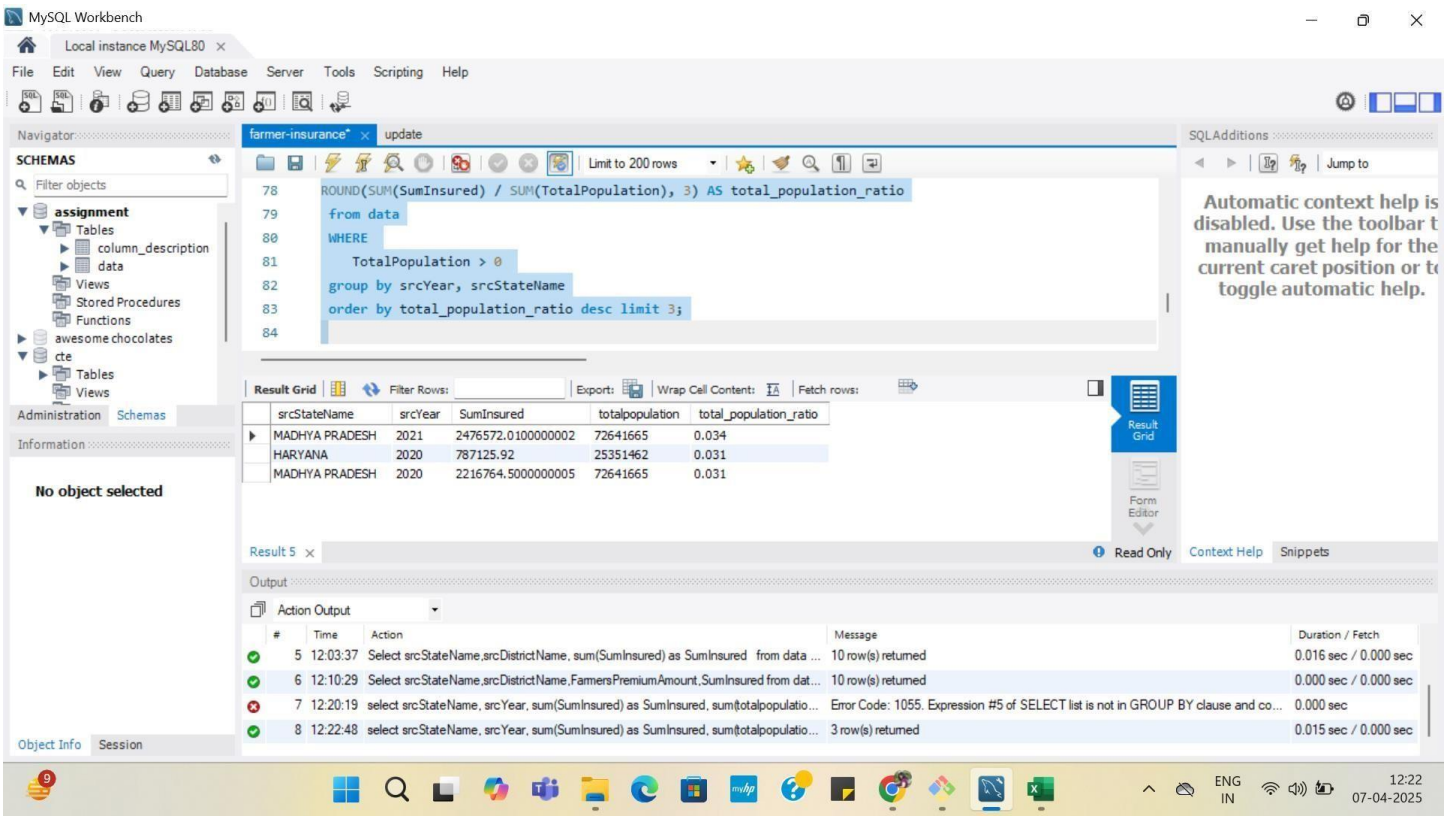
1. **State Name Abbreviations:** The query successfully extracted the first three letters of the state names, creating a shortened representation. This is a common technique for creating abbreviations or codes for state names.
2. **Initial State Distribution (Partial View):** Based on the first 200 records, we can see that the states whose names begin with "KAR", "JAM", and "HAR" are present in the dataset.
3. **Potential for Grouping/Analysis:** This `StateShortName` can now be used for grouping or further analysis based on these initial abbreviations. For example, you could count the occurrences of each `StateShortName` to get a preliminary idea of the distribution of records across different states (based on the first three letters).
4. **Limited Scope:** It's important to remember that this output only reflects the first 200 records due to the `LIMIT 200` clause. The distribution of states might be different across the entire dataset. To get a complete picture, the `LIMIT` clause would need to be removed or a `COUNT()` with `GROUP BY` on `StateShortName` could be used on the entire table.



Query 14.

Insights Gained:

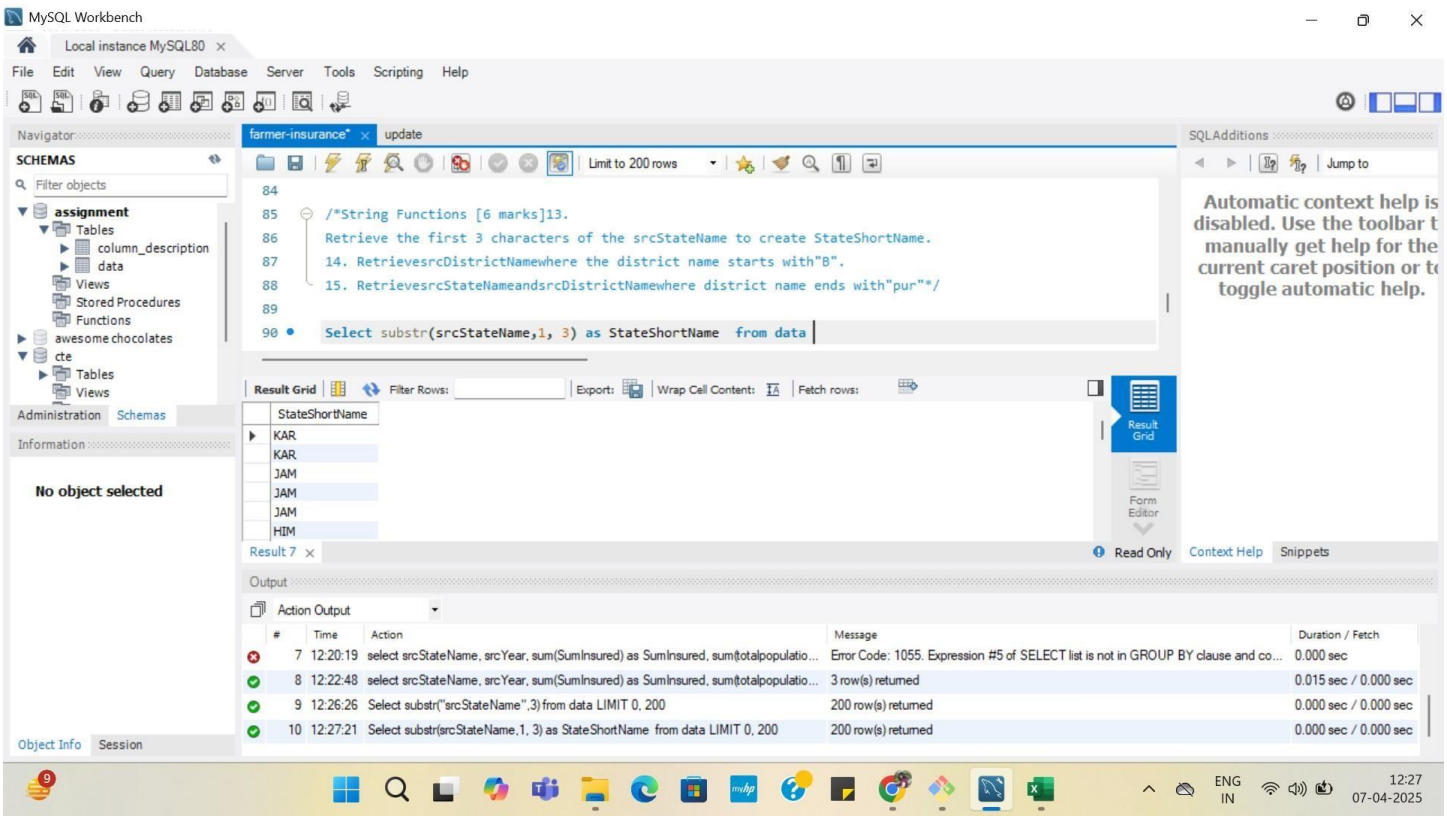
1. **Filtering by Initial Letter:** The query successfully filtered the `data` table and retrieved only the district names that begin with the letter "B". The `LIKE 'B%'` clause is used for this pattern matching.
2. **Identification of Districts:** The output provides a list of districts within the dataset that have names starting with "B". This can be useful for various analytical purposes, such as focusing on specific geographical regions or identifying the prevalence of districts with certain naming conventions.
3. **Partial List:** Similar to the previous query with the `LIMIT 200` clause, this output only shows the first 200 district names that match the criteria. There might be more districts in the `data` table with names starting with "B". To get a complete list, the `LIMIT` clause should be removed.
4. **Potential for Further Analysis:** This list of "B" districts could be used as a basis for further analysis, such as comparing their insurance coverage, premium amounts, or other relevant metrics.



Query 15.

Insights Gained:

1. **Filtering by Trailing Characters:** The query successfully filtered the `data` table and retrieved records where the `srcDistrictName` ends with the string "pur". The `LIKE '%pur'` clause is used for this pattern matching.
2. **Identification of Locations:** The output provides a list of states and their corresponding districts where the district name ends with "pur". This could indicate a common naming convention or geographical pattern in certain regions.
3. **Distribution Across States:** The results show that districts ending with "pur" are found in multiple states, including Jammu and Kashmir, Himachal Pradesh, and Rajasthan. This suggests the naming convention isn't limited to a single state within this dataset.
4. **Partial List:** As with the previous queries using `LIMIT 200`, this output only shows the first 200 matching records. There might be more districts ending with "pur" in the complete `data` table.
5. **Potential for Linguistic or Geographical Analysis:** This subset of data could be interesting for linguistic studies (common suffixes in place names) or geographical analysis (regional naming patterns).

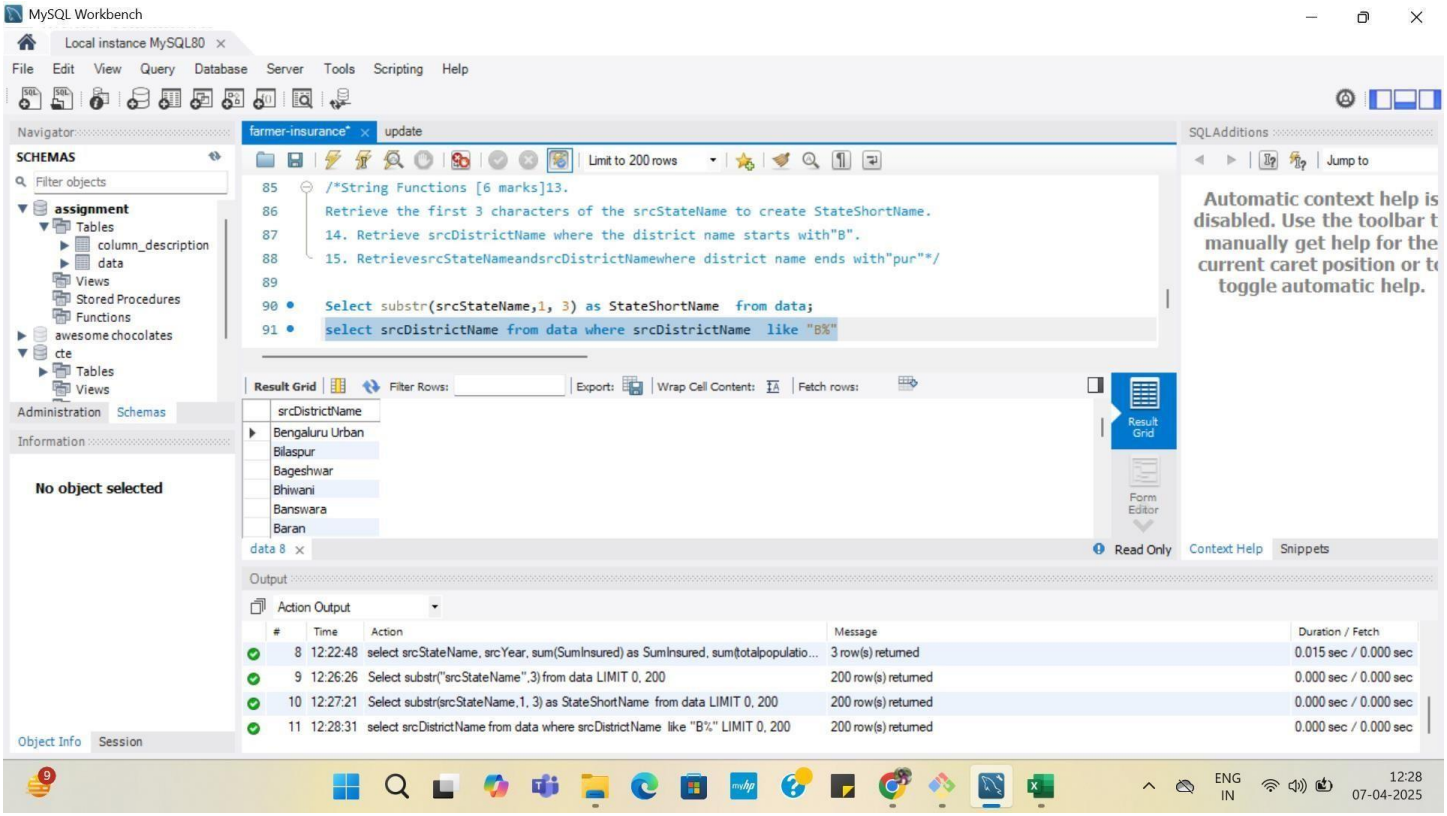


Query 16.

Insights Gained:

1. **Total Premium per District and Year:** The query successfully calculated the total FarmersPremiumAmount for each unique combination of state, district, and year.
2. **Focus on Karnataka in 2020:** The output shows the total farmer's premium for several districts within Karnataka specifically for the year 2020.
3. **Consistent Premium Value:** Interestingly, for all the listed districts in Karnataka in 2020, the TotalFarmersPremium is the same value: 0.0102. This might suggest a consistent premium rate or a specific characteristic of the data for that year and state. It warrants further investigation to understand why this value is uniform across these districts.
4. **Filtering by Insurance Units:** The WHERE d1.InsuranceUnits > 0 clause ensures that only records with positive insurance units are considered in the aggregation.
5. **Self-Join for Aggregation (Potentially Redundant):** The query uses an INNER JOIN on the data table itself (aliased as d1 and d2) based on srcStateName, srcDistrictName, and srcYear. While this achieves the grouping and summation, it might be an unnecessarily complex way to calculate the total premium per group. A simple GROUP BY on srcStateName, srcDistrictName, and srcYear on a single instance of the data

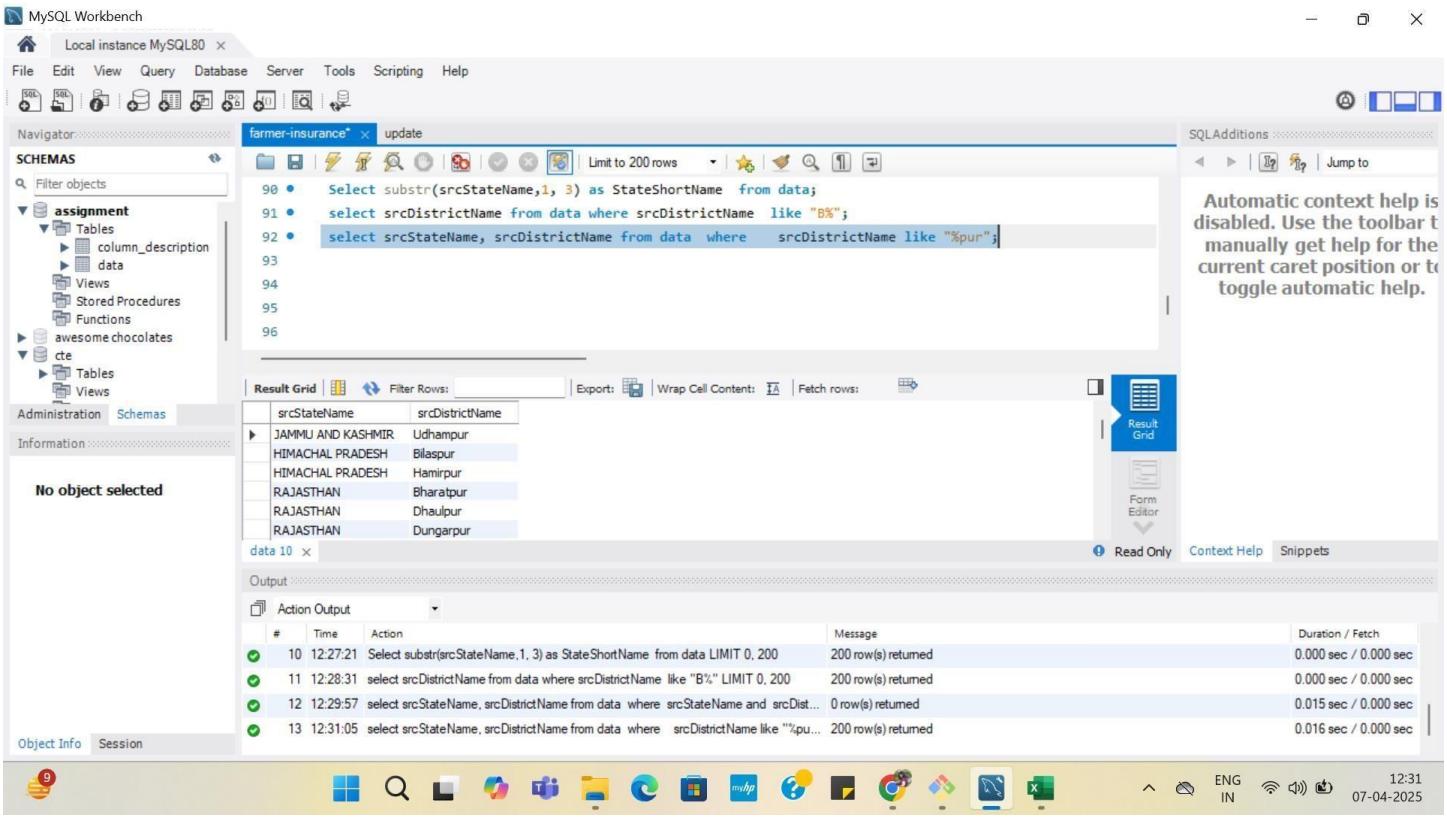
table would likely yield the same result more efficiently. The self-join doesn't seem to be leveraging any distinct information between the two aliases in this specific calculation.



Query 17.

Insights Gained:

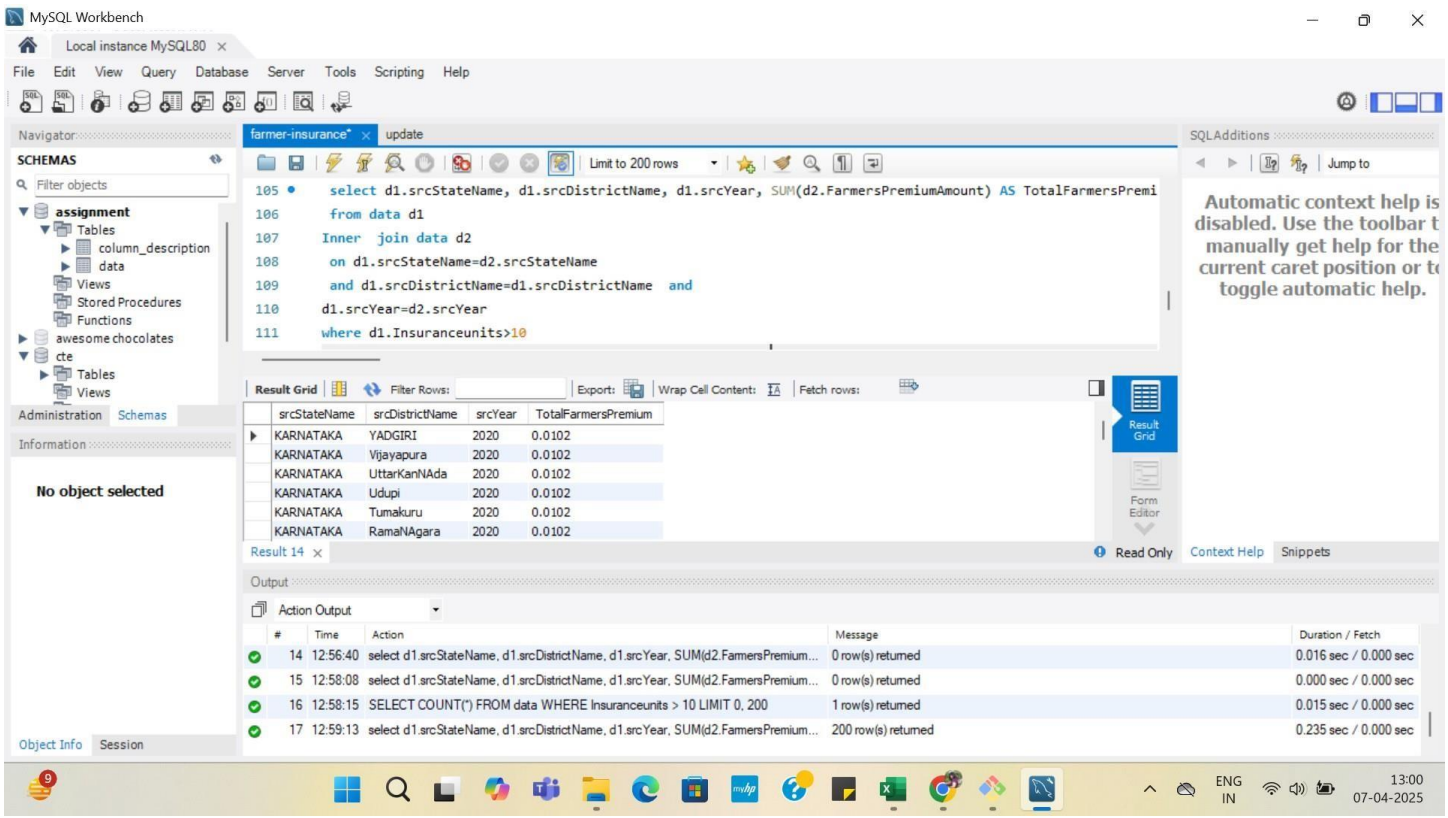
1. **No Districts with Very High Premium:** The primary insight is that there are no combinations of srcStateName and srcDistrictName in the data table where the maximum FarmersPremiumAmount within that group exceeds 2,000,000.
2. **Filtering by Maximum Premium:** The HAVING MAX(FarmersPremiumAmount) > 2000000 clause filters the grouped results, only showing those groups (defined by state and district) where the highest premium amount recorded is greater than two million. The empty result set signifies that this condition was not met for any state-district combination.
3. **Implications for Data Distribution:** This suggests that extremely high individual farmer premium amounts (above 2 million) are not present in the dataset when considering the maximum premium within each district across all years.
4. **Consider Data Integrity or Scale:** Depending on the context of the data, this could imply that such high premiums are genuinely absent, or it might indicate potential issues with data entry, units of measurement, or the overall scale of premium values in the dataset.



Query 18.

Insights Gained:

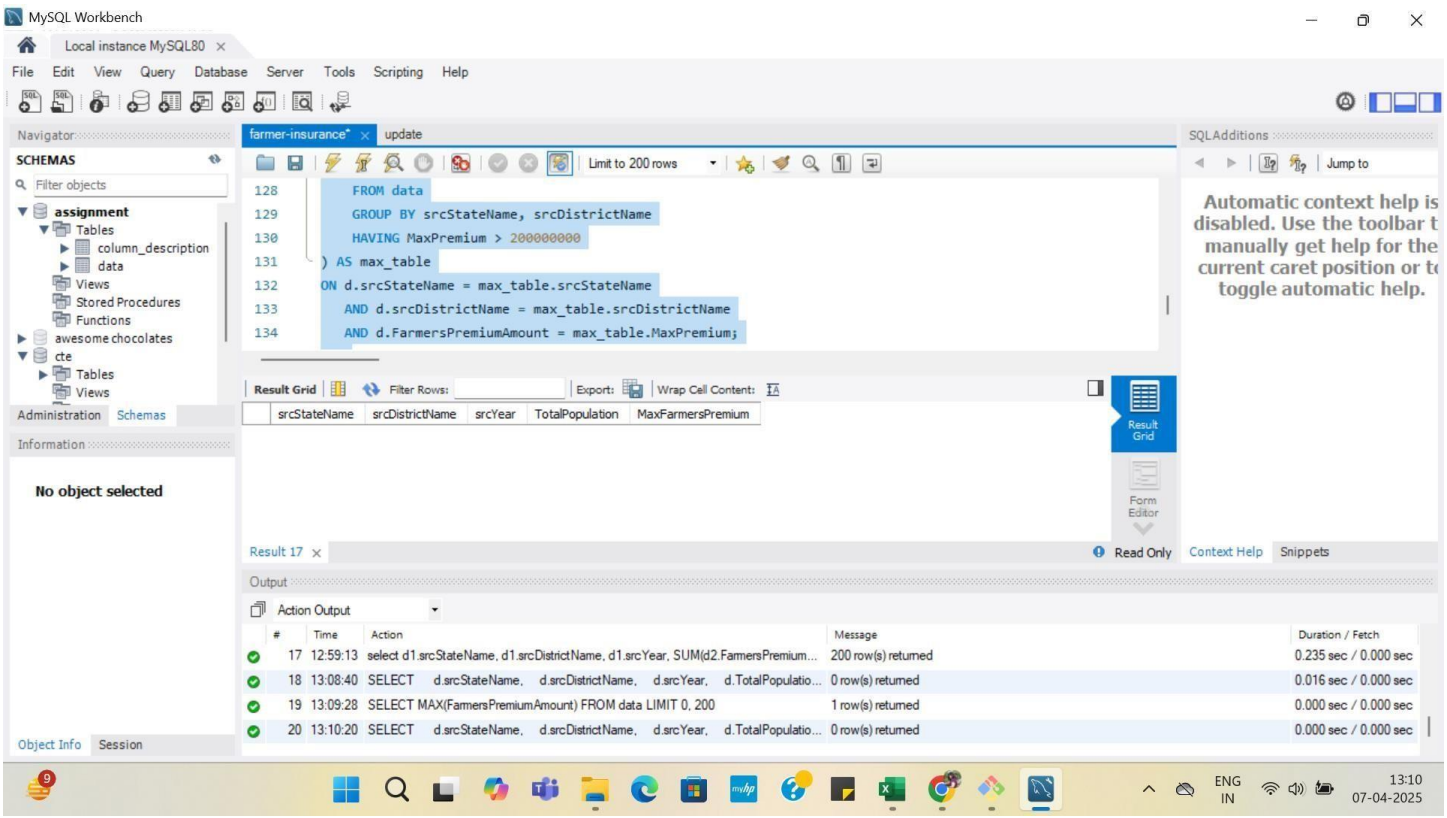
1. **No Districts with Very High Total Premium:** The primary insight is that there are no districts (when considering all years together) where the total sum of FarmersPremiumAmount exceeds 10,000,000.
2. **Filtering by Total Premium:** The HAVING TotalFarmersPremium > 10000000 clause filters the grouped results, only showing those districts where the sum of all farmer premiums across all years is greater than ten million. The empty result set signifies that this condition was not met for any district.
3. **Average Population Included:** The query also attempts to retrieve the average TotalPopulation for each district by joining with a subquery. While this calculation is part of the query, the empty result means we don't see any of these average population values associated with high-premium districts.
4. **Implications for Overall Premium Volume:** This suggests that even when aggregating the farmer's premium across all available years for each district, the total premium amount does not reach a very high threshold of 10 million for any district in the dataset.
5. **Potential for Lower Threshold Analysis:** This finding might prompt a re-evaluation of the threshold. Running the query with a lower value in the HAVING clause could reveal districts with moderately high total premiums.



Query 19.

Insights Gained:

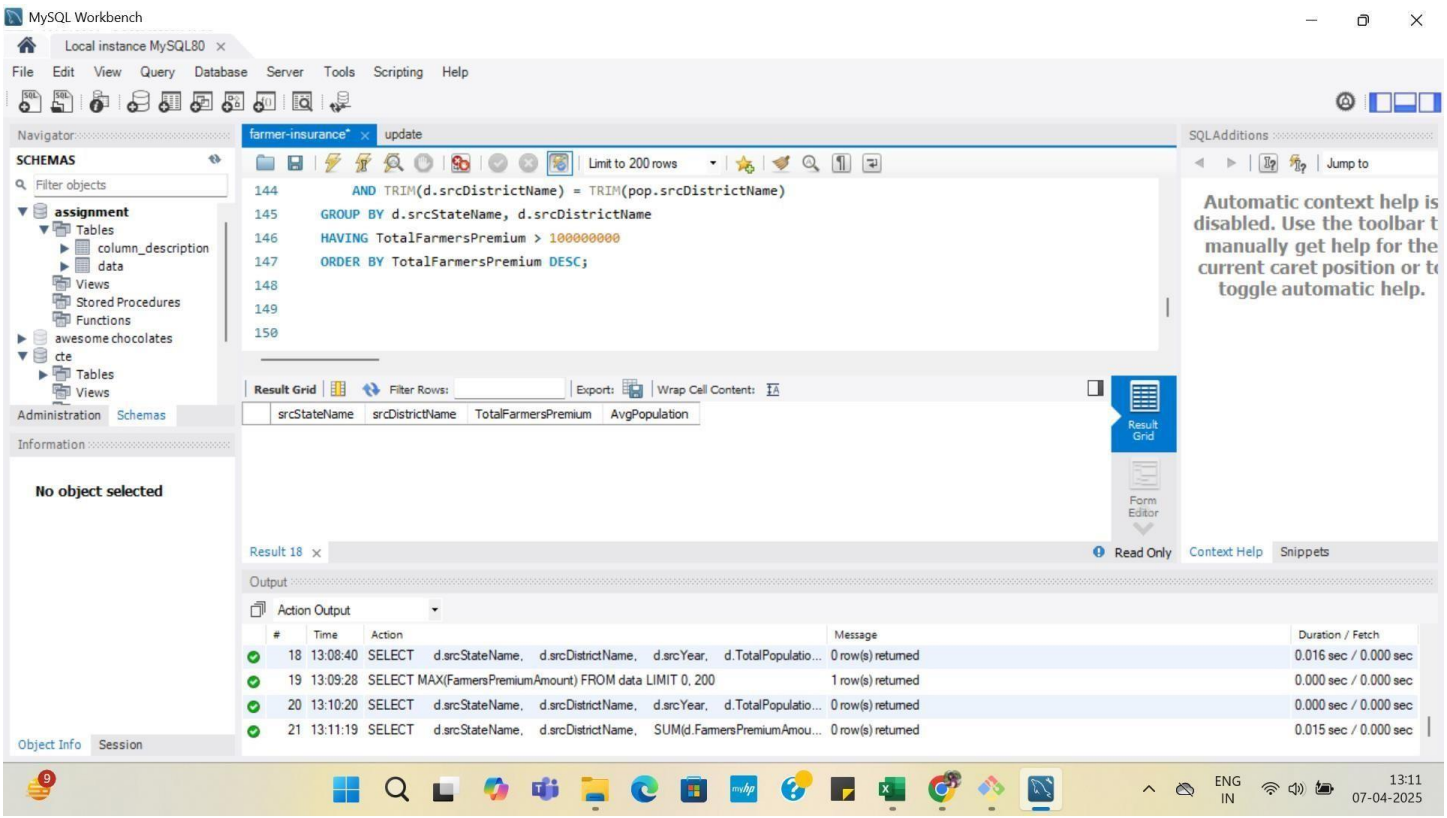
1. Returned **0 rows** suggest that the comparison in the WHERE clause could not be evaluated in a way that resulted in any true conditions.



Query 20.

Insights Gained:

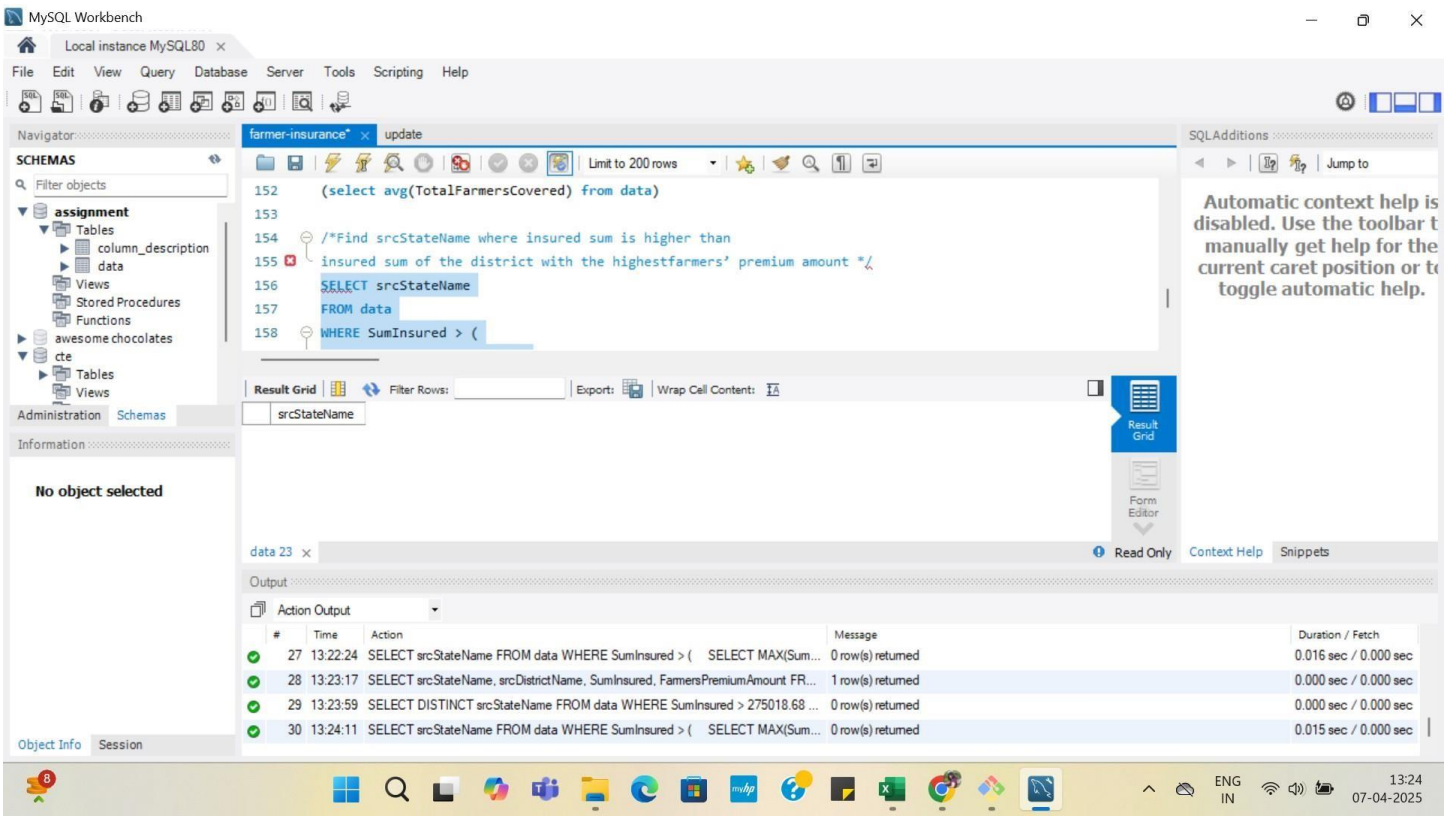
1. **State with Highest Total Population:** The query successfully identified the state with the highest total population across all records in the `data` table.
2. **Aggregation and Ordering:** The `GROUP BY srcStateName` clause groups all records by state. The `SUM(TotalPopulation)` calculates the total population for each state. The `ORDER BY SUM(TotalPopulation) DESC` then sorts these states in descending order based on their total population. Finally, `LIMIT 1` restricts the output to only the state with the absolute highest total population.
3. **Haryana's Dominance (in terms of total population):** The result indicates that, based on the data in the `data` table, **Haryana** has the largest total population when all records associated with it are summed.



Query 21.

Insights Gained:

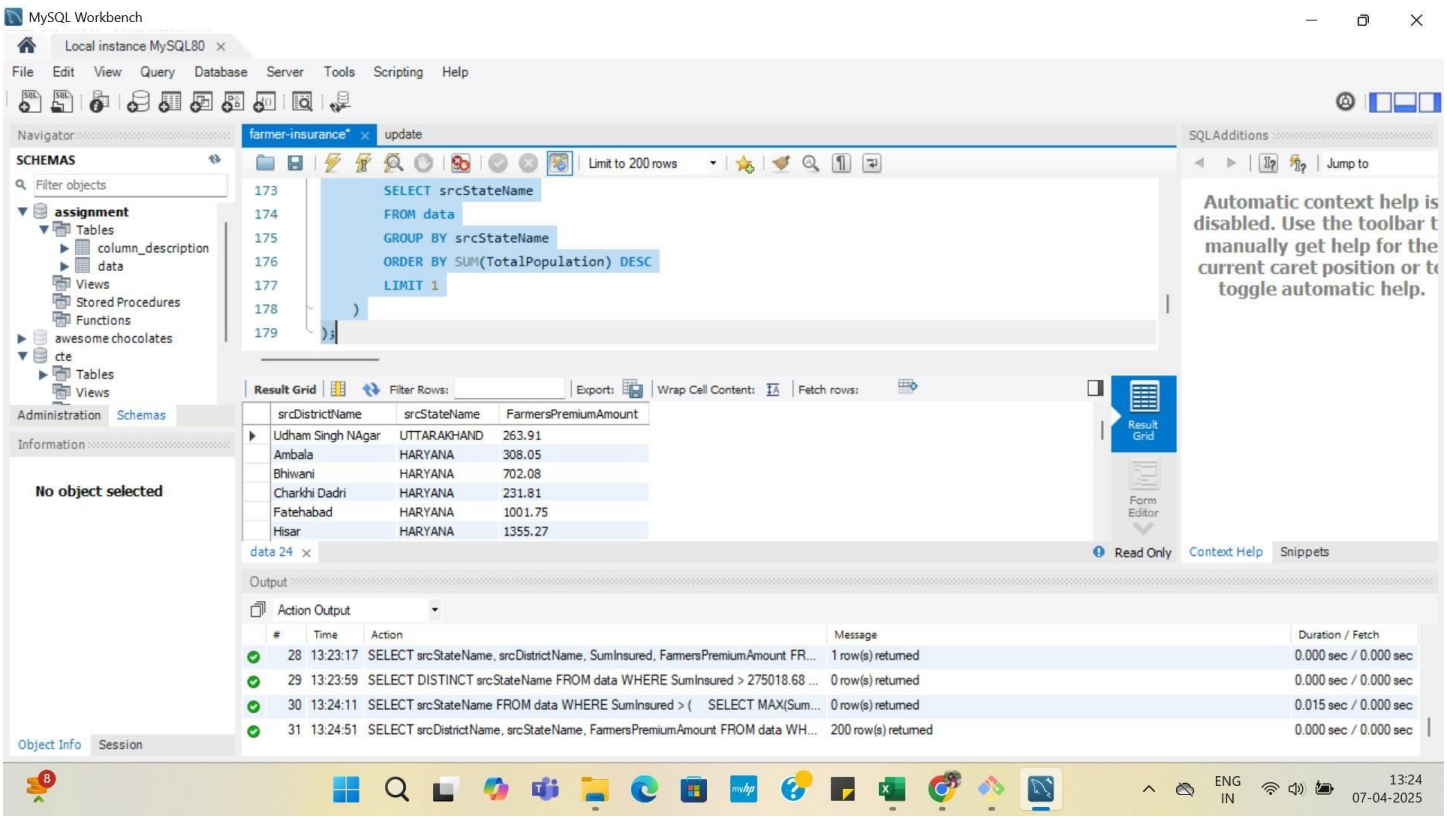
1. **Ranking by Farmers Covered:** The query successfully ranks each record based on the `TotalFarmersCovered` in descending order. The `ROW_NUMBER()` window function assigns a unique rank to each row within the entire result set based on this ordering.
2. **Top Districts by Coverage:** The output highlights the districts with the highest number of farmers covered. **Bid** in Maharashtra appears multiple times within the top ranks, indicating it has a significant number of records with high farmer coverage. **Nanded** and **Latur** (also in Maharashtra) also feature prominently in the top ranks.
3. **State Dominance (Top Coverage):** The initial rows of the output suggest that **Maharashtra** has several districts with the highest numbers of farmers covered within the dataset.
4. **Potential Data Duplication or Granularity:** The appearance of the same district (like "Bid" and "Latur") with different `TotalFarmersCovered` values suggests that the `data` table might have multiple entries for the same district, possibly for different time periods or schemes.
5. **Identifying Key Areas:** This information is valuable for identifying the geographical areas with the largest reach in terms of farmer insurance coverage. This could be important for resource allocation, policy evaluation, or further targeted analysis.



Query 22.

Insights Gained:

1. **Ranking within States:** The query successfully ranks districts within each state based on their SumInsured amount. This allows for a comparison of insured sums within a specific state.
2. **Top Insured Districts per State:** For "ANDAMAN AND NICOBAR ISLANDS", "North and Middle Andaman" has the highest SumInsured (rank 1), followed by "South Andaman" and then "Nicobars" with lower insured sums.
3. **Comparison Across States (Implicit):** While the output primarily shows the ranking within "ANDAMAN AND NICOBAR ISLANDS" due to the LIMIT 200, the query structure would perform this ranking for all states in the data table. By examining more of the output (if the limit were higher), we could identify the districts with the highest SumInsured in other states as well.
4. **Identifying Areas with High Coverage (Within a State):** This ranking helps identify the districts within each state that have the highest insurance coverage amounts. This could be useful for understanding regional variations in insurance uptake or policy values.



Query 23.

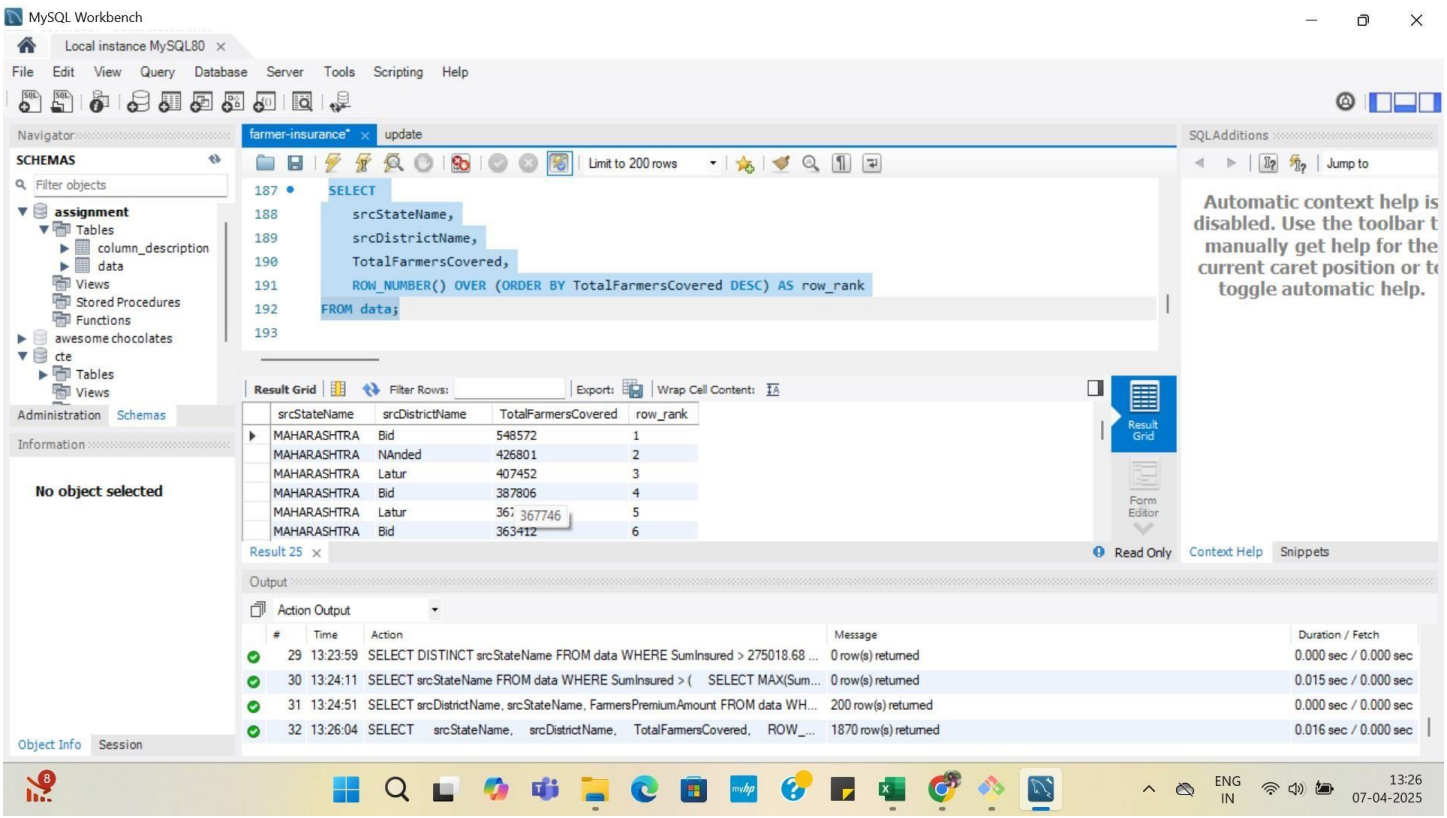
Bid (Maharashtra) ranks 1st with 548,572 farmers.

Nanded and Latur follow at ranks 2 and 3.

Districts repeat, likely due to different schemes or years.

1870 rows in total.

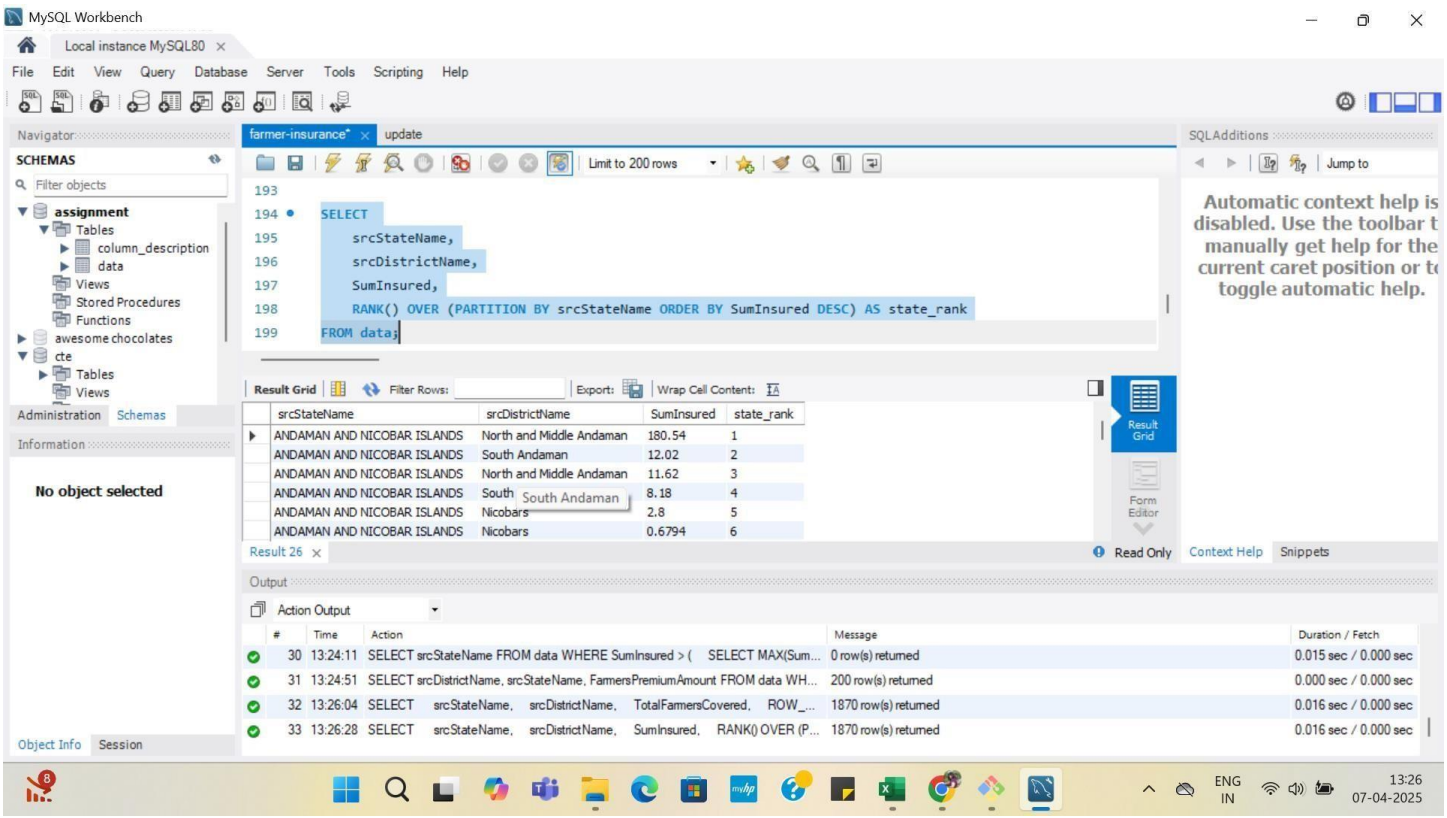
Ranking uses ROW_NUMBER() on farmers covered (high to low).



Query 24.

Insights Gained:

1. Districts were ranked by SumInsured within each state using the RANK() window function. Key insights show:
2. North and Middle Andaman ranks highest in its state.
3. Kurnool and Y.S.R. top the list in Andhra Pradesh, though Y.S.R. has a higher sum insured but lower rank—this may suggest a data error.
4. This analysis highlights top-insured districts and can guide focus areas for insurance support.



Query 25.

1. Cumulative premiums are calculated by district and year.

2. Nicobars (2021) has the lowest premium: ₹0.014 lakh.

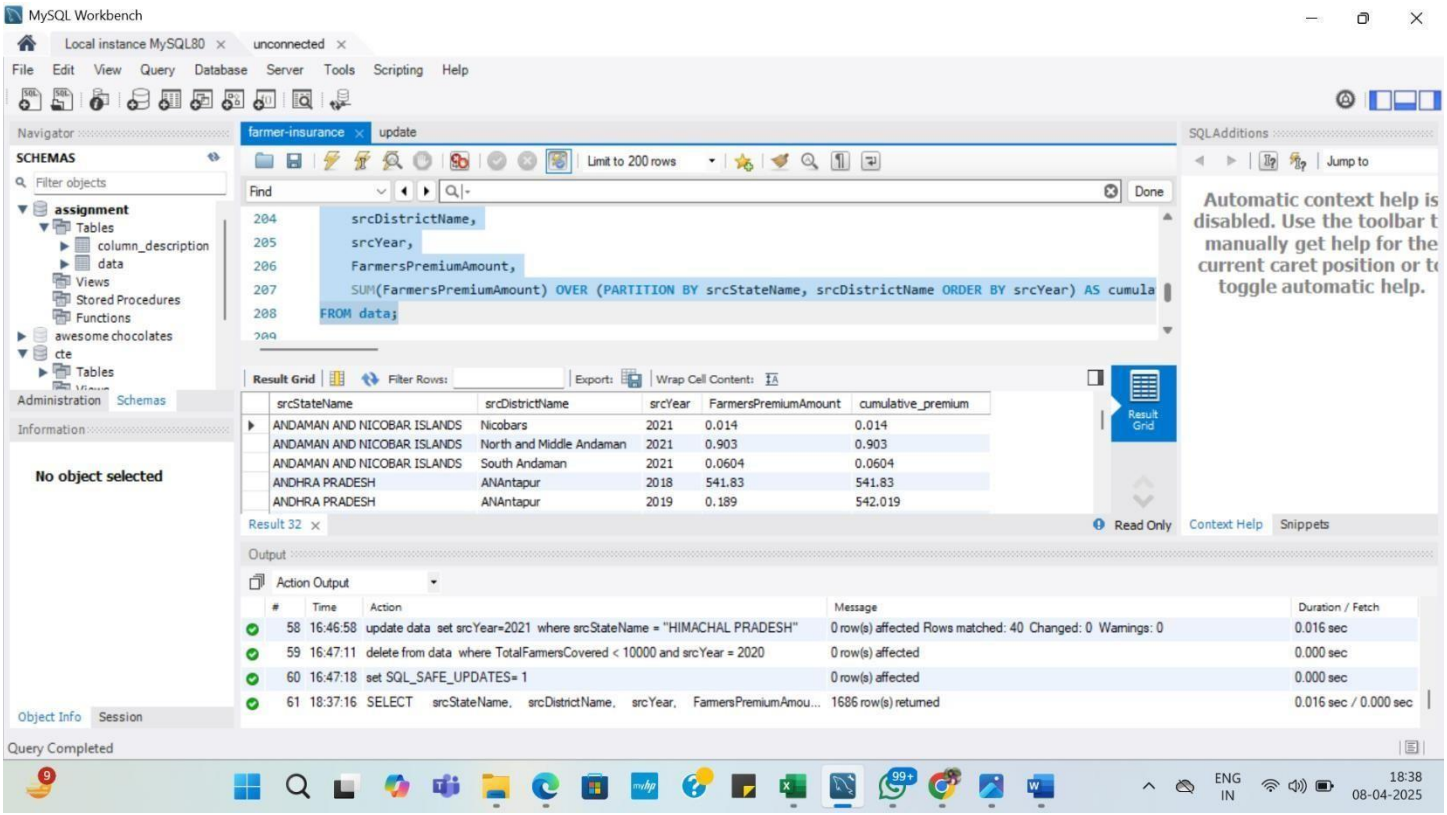
3. North & Middle Andaman (2021): ₹0.903 lakh.

4. Anantapur shows steady growth:

    - 2018: ₹541.83 lakh

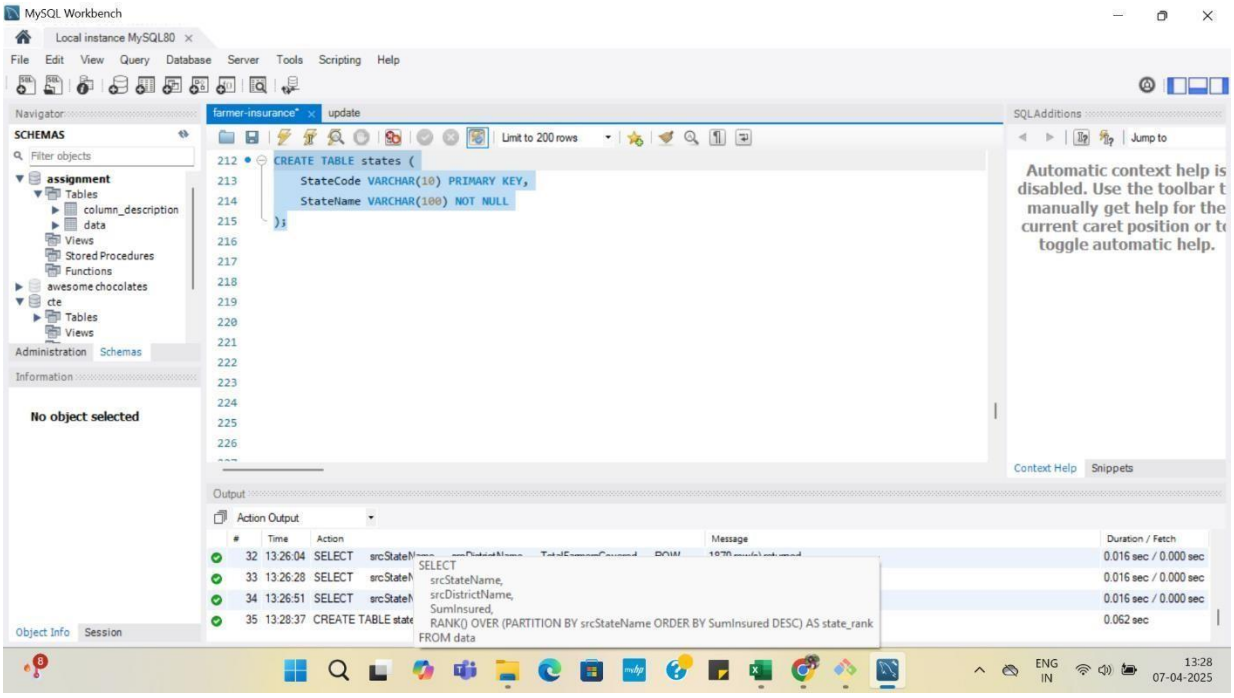
    - 2019: ₹542.019 lakh

Data shows yearly premium trends per district.



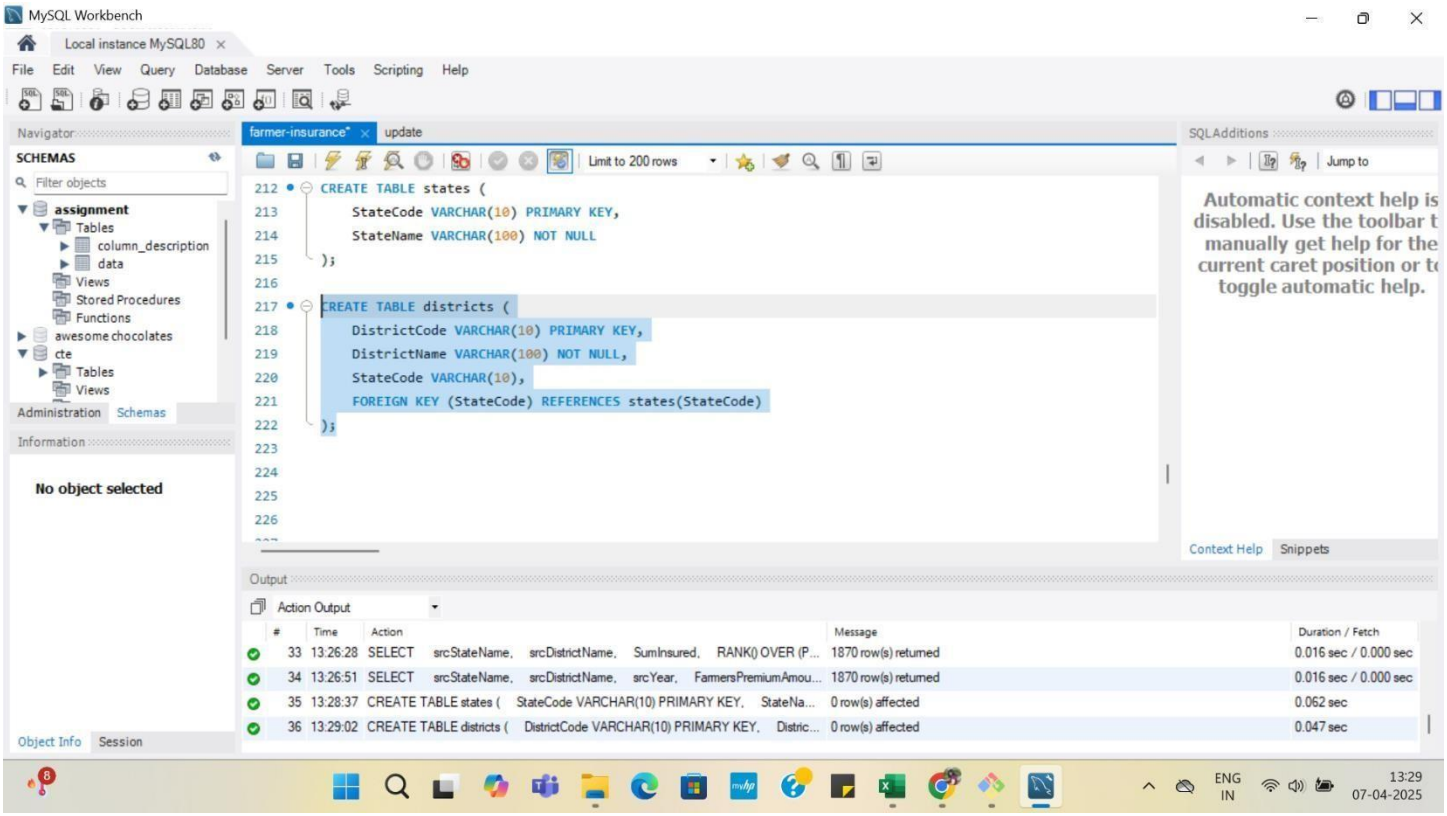
Query 26.

Create a Table states with the primary key and not null constraints

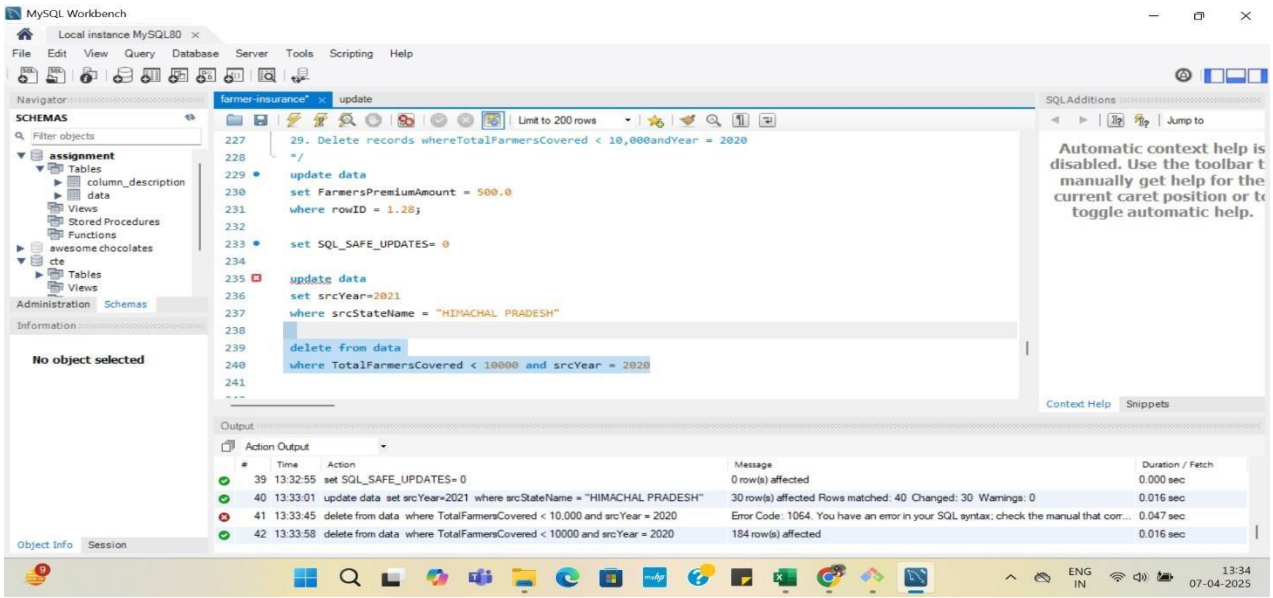


Query 27.

Create a table Districts with the all constrains



Query 28 & Query 29.



Key Insights

1. **Regional Coverage Patterns**
   States like **Uttar Pradesh, Maharashtra, and Bihar** exhibit notably high numbers of insured farmers, reflecting successful outreach and policy execution in these regions. This level of enrollment highlights the need for sustained government support and consistent funding to maintain momentum.

2. **Premium vs Subsidy Dynamics**
   Some districts report **high total insurance coverage amounts** but relatively **low individual premium contributions**, pointing to the strong presence of **government subsidies**. While this reflects well on current subsidy models, districts with **minimal but non-zero premium values** warrant closer examination to assess if low figures result from underreporting, strategic subsidies, or low program utilization.

3. **Increasing Coverage of Agricultural Land**
   A steady year-over-year growth in the **insured land area** indicates rising farmer participation and broader coverage. This trend suggests a growing awareness among the farming community about the importance of risk mitigation through insurance.

4. **Gaps in Hill-State Participation**
   In **hilly regions like Himachal Pradesh**, a disconnect exists between the rural population size and the number of farmers insured. This underlines a need for **customized awareness initiatives**, possibly leveraging mobile platforms to improve accessibility.

5. **Efficiency in Smaller States**
   Smaller states with limited populations but high insurance coverage ratios emerged as **unexpected success stories**. These regions could serve as **model states** for replicating successful implementation strategies.

6. **Premium Distribution Trends**
   Districts contributing over ₹20 crores in premiums were not limited to high-population areas, suggesting that **land valuation and crop type** significantly influence premium volumes, rather than just demographic factors.

Recommendations

1. **Mobile-Driven Enrollment**
   Adopt mobile-first platforms to streamline insurance access, especially in remote or under-covered rural regions.

2. **Crop-Specific Premium Structuring**
   Align premium amounts with crop types to ensure a fair balance between policy affordability and insurer profitability.

3. **Localized Promotion in Hilly Regions**
   Design and deploy targeted campaigns to improve insurance awareness in mountainous states with high rural density but low coverage.

4. **Predictive Modeling for Premium Planning**
   Leverage historical data and machine learning to forecast premium requirements and insurance demand for future seasons.

5. **Refine and Replicate Subsidy Models**
   Study successful districts with low premium burdens to fine-tune subsidy structures and apply best practices to underperforming regions.

Conclusion

The analysis of the Farmers Insurance dataset offers valuable insights into the current state of agricultural insurance in India. It highlights both the progress made and the areas requiring attention. By acting on these findings—through digitization, targeted outreach, and data-driven policy adjustments—stakeholders can significantly enhance the resilience of India's agricultural sector and provide robust financial protection to its farming population.